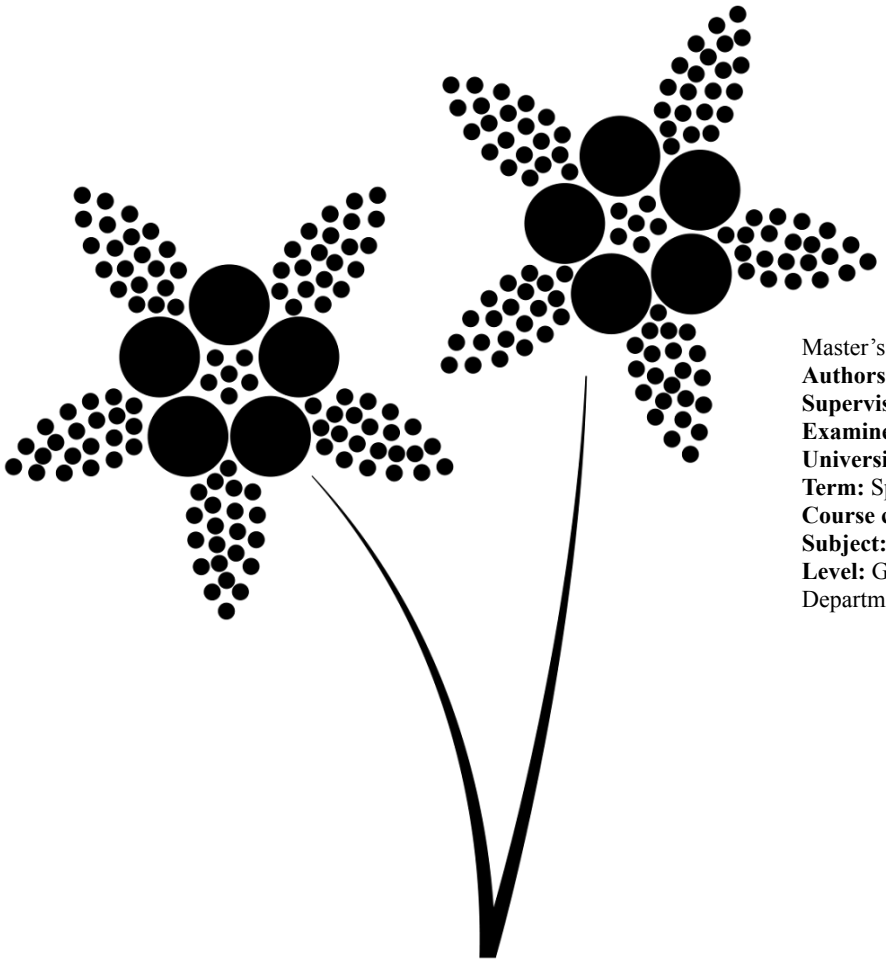
Linnæus University

Accessibility Barriers Across Swedish Public-Sector Websites

A Taxonomy-Driven Analysis



Master's thesis
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Abstract

This study examines accessibility barriers in Swedish public-sector websites using a taxonomy-driven, cross-domain analytical approach. As digital public services increasingly function as primary access points for citizens, equitable and inclusive access has emerged as a critical societal and institutional concern. Although recognized standards such as the Web Content Accessibility Guidelines (WCAG) and regulatory frameworks exist in the European Union and Sweden, previous research has identified ongoing accessibility issues across public-sector platforms.

To address these limitations in existing research, this study employs a structured methodology that combines automated accessibility assessment with taxonomy-based normalization and persona-informed interpretation. Thirty-six websites across three domains—universities, municipalities, and healthcare organizations—were analyzed using automated evaluation tools. The resulting data were transformed into standardized issue definitions, categorized by barrier type, and mapped to user-impact groups and interpretive personas.

The findings indicate that accessibility barriers are common but unevenly distributed across domains. A total of 4,469 issue instances were identified, with universities and municipalities reporting more accessibility issues compared to healthcare websites. Accessibility issues are concentrated in structural and visual categories, particularly structure and semantics, visual design and contrast, and compatibility-related issues. These barriers primarily affect users' ability to navigate content and understand information, as reflected in the significant impact on assistive-technology navigation and reading clarity personas.

The study further demonstrates that accessibility barrier profiles differ systematically across domains, underscoring the importance of context-sensitive analysis. By integrating technical detection with structured interpretation, this research provides a scalable and interpretable framework for analyzing accessibility barriers beyond compliance metrics. The findings support a transition toward data-driven, user-oriented accessibility practices in public-sector digital services.

Keywords

Web accessibility; WCAG; public-sector websites; accessibility evaluation; accessibility barriers; taxonomy framework; personas; comparative analysis; automated accessibility testing; digital accessibility.

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1. Introduction **1**
1.1. Background and problem 1
1.2. Topic justification 2
1.3. Purpose statement and research questions 2
1.4. Scope and Limitations 3
1.5. Thesis organization 4
2. Literature review **5**
2.1. Web Accessibility and Public Sector Context 5
2.2. WCAG 2.2 and Accessibility Compliance 5
2.3. Automated Accessibility Evaluation in Prior Studies 6
2.4. Taxonomy and Structured Accessibility Classification 7
2.5. Accessibility Barriers in Public Sector Websites 8
2.6. Persona-Informed Accessibility Research 8
2.7. Research Gap and Chapter Summary 9
3. Theoretical framework **10**
3.1. Conceptual Foundations 10
3.2. Accessibility As Barrier-Oriented Analysis 11
3.3. Taxonomy Framework 11
3.3.1. Taxonomy Structure 11
3.3.2. Mapping Logic 13
3.3.3. Relationship to WCAG 14
3.3.4. Theoretical Role of the Taxonomy 14
3.3.5. Design Considerations and Boundaries 14
3.3.6. Summary 15
3.4. Persona Framework 15
3.4.1. Operationalization of Personas 15
3.4.2. Persona Definitions 16
3.4.2.1. Assistive Technology Navigation Persona 17
3.4.2.2 Reading Clarity Persona 18
3.4.2.3. Task Guidance Persona 18
3.4.3. Role in the Analytical Model 19
3.4.4. Summary 19
3.5. Domain Comparison Framework 20
3.6. Analytical Model 20
3.6.1. Analytical Pipeline 21
3.6.2. Units of Analysis 22
3.6.3. Model Boundaries and Assumptions 23
3.7. Summary 23
4. Methodology **24**
4.1. Philosophical tradition 24
4.2. Methodological approach 24
4.3. Methods/techniques for data collection 24
4.3.1. Study Context And Sampling Strategy 24

4.3.2. Data Collection Pipeline 25
4.3.3. Evaluation Configuration 26
4.4. Methods/techniques for data analysis 26
4.4.1. Measures And Variables 26
4.4.2. Data Processing And Normalization 27
4.4.3. Analytical Procedure 28
4.5. Reliability, Validity And Reproducibility 28
4.6. Ethical considerations 29
5. Empirical findings 30
5.1. Overview of the Dataset 30
5.2. RQ1: Extent of Accessibility Barriers 30
5.2.1. Overall Burden 31
5.2.2. Domain Distribution 31
5.2.3. Diversity 32
5.2.4. Severity distribution 32
5.2.5. Normalized Accessibility Burden 33
5.2.6. Summary 34
5.3. RQ2: Types of Accessibility Barriers and Persona Interpretation 34
5.3.1. Distribution of Accessibility Issue Categories 34
5.3.2. Impact Group Distribution 35
5.3.3. Persona-Based Interpretation 36
5.4. RQ3: Comparative Analysis of Accessibility Barriers Across Domains 37
5.4.1. Category Distribution Across Domains 37
5.4.2. Persona Distribution Across Domains 38
5.4.3. Summary of Domain Differences 39
6. Discussion 41
6.1. Extent of Accessibility Barriers 41
6.2. Accessibility Barrier Types and User Impact 42
6.3. Cross-Domain Differences in Accessibility Barriers 43
6.4. Methodological Reflections 44
6.5. Practical Implications and Recommendations 45
6.5.1. General Accessibility Recommendations 45
6.5.2. Study-Specific Recommendations 46
7. Conclusion 48
7.1. Conclusions 48
7.2. Contributions 49
7.3. Limitations 50
7.4. Future research 51
References 52

Appendices

Appendix A: Report on the use of generative AI in master thesis
Appendix B: Description of the work division between the authors
Appendix C: Mapping of Issue Definitions to Issue Categories and Impact Groups
Appendix D: Evaluated Websites by Domain and Page Coverage
Appendix E: Accessibility issue matrix

1. Introduction

1.1. Background and Problem

Digital public services have become a primary means through which citizens interact with public institutions such as universities, municipalities, and healthcare organizations. Within the broader development of e-government, websites and online platforms now function as essential infrastructures for accessing information, services, and administrative processes (Mergel, Edelmann, & Haug, 2019). While this digital transformation has improved efficiency and accessibility for many users, it has also raised critical concerns regarding inclusion and equitable access.

Not all users are equally able to benefit from digital services. Individuals with disabilities, limited digital literacy, or restricted access to technology may encounter significant barriers when interacting with public-sector websites (Van Deursen & Helsper, 2015; Johansson, 2021). Digital accessibility therefore plays a central role in ensuring that online services can be perceived, understood, navigated, and used by diverse user groups (Henry, Abou-Zahra, & Brewer, 2014). When accessibility barriers are present, they can lead to exclusion from essential services and contribute to broader social inequalities (Lazar, Goldstein, & Taylor, 2015).

To address these challenges, international standards such as the Web Content Accessibility Guidelines (WCAG), developed by the World Wide Web Consortium (W3C), provide structured criteria for accessible design (W3C, 2024). These guidelines are widely adopted in policy and legislation, including the European Union's Web Accessibility Directive (EU) 2016/2102 and its implementation in Sweden through the Act on Accessibility to Digital Public Services (2018:1937). Despite these regulatory frameworks, research consistently shows that many public-sector websites continue to exhibit accessibility barriers, including issues related to content structure, navigation, and compatibility with assistive technologies (Alajarmeh, 2021; Vollenwyder et al., 2023). Studies in the Swedish context further indicate uneven compliance and persistent accessibility challenges across institutions (Jonsson, Gustavsson, & Gulliksen, 2023).

Existing research has frequently relied on automated tools to evaluate accessibility by identifying technical violations of WCAG guidelines (Alajarmeh, 2021; Campoverde-Molina et al., 2023; Inal et al., 2022; Valtolina & Fratus, 2022). While these approaches are effective for detecting issues, they often result in descriptive lists of problems and are commonly applied to individual organizations or single domains. As a result, there is limited understanding of how accessibility barriers are structured across different types of public institutions, how recurring barrier patterns differ between domains, and how such patterns relate to user-impact considerations. Furthermore, automated evaluation methods have known limitations, as they cannot capture all accessibility issues and may miss context-dependent or interaction-based barriers (Vigo, Brown, & Conway, 2013).

This creates a need for approaches that move beyond isolated compliance-oriented evaluations toward structured, comparative, and interpretable analysis of accessibility

barriers. In particular, there is a lack of cross-domain studies that apply consistent evaluation logic across multiple types of public institutions while also connecting technical findings to user-impact perspectives. Addressing this gap is essential for developing a more comprehensive understanding of accessibility challenges in public-sector digital services.

1.2. Topic Justification

The proposed study is relevant from both societal and academic perspectives. From a societal perspective, accessible digital public services are essential for ensuring equitable access to information and services. Public-sector websites serve diverse user groups, including individuals with disabilities and varying levels of digital literacy. When accessibility barriers are present, users may be excluded from essential services such as healthcare information, educational resources, and administrative processes. Improving accessibility is therefore directly linked to inclusion, equal participation, and the effective functioning of digital public services (Lazar, Goldstein, & Taylor, 2015; Van Deursen & Helsper, 2015).

From an academic perspective, the study contributes to Information Systems research by addressing limitations in existing approaches to web accessibility evaluation. Prior research has often focused on compliance measurement or technical issue detection, frequently within single organizational contexts (Alajarmeh, 2021; Vollenwyder et al., 2023). While such approaches are useful for identifying violations, they provide limited support for understanding how accessibility barriers are structured, how they differ across institutional domains, and how they relate to user-impact considerations.

This study contributes by adopting a barrier-oriented and taxonomy-driven analytical approach that enables structured cross-domain comparison and interpretation. By combining automated accessibility evaluation with normalized issue classification and literature-informed persona interpretation, the study extends how accessibility data can be analyzed and translated into meaningful insights. In addition, the study supports the development of data-informed recommendations that can assist public organizations in prioritizing and addressing accessibility barriers in a systematic and context-aware manner.

1.3. Purpose Statement and Research Questions

The purpose of this thesis is to analyze accessibility barriers across selected Swedish public institution websites and to examine how these barriers are distributed, structured, and interpreted across different public-service domains. The study focuses on three domains: universities, municipalities, and healthcare organizations in order to enable systematic cross-domain comparison under a consistent evaluation and analytical framework.

The study uses an automated evaluation pipeline supported by a custom analytical platform to ensure consistency in data collection and structuring. Rather than treating

accessibility evaluation as a compliance or auditing task, the thesis adopts a barrier-oriented and taxonomy-driven approach that enables structured comparison and supports interpretation of user-impact implications. To strengthen interpretability, findings are organized through an issue taxonomy and interpreted using literature-informed personas, linking technical accessibility issues to likely patterns of user impact.
In addition, the study aims to derive data-informed recommendations to support accessibility improvement efforts in public-sector digital services.

The study is guided by the following research questions:

RQ1: To what extent do selected Swedish public institution websites exhibit WCAG-aligned accessibility barriers based on automated evaluation?

RQ2: Which types of accessibility barriers are most prevalent within and across these websites, and how can these be interpreted through literature-informed personas?

RQ3: How do accessibility barrier profiles differ between universities, municipalities, and healthcare websites?

1.4. Scope and Limitations

The study focuses on a selected sample of Swedish public institution websites across three domains: universities, municipalities, and healthcare organizations. The scope is limited to publicly accessible web pages within a defined page selection strategy, and the findings are therefore representative of the studied sample rather than all public-sector websites.

Accessibility evaluation is conducted using automated tools. While these tools are effective for identifying detectable and measurable accessibility issues, they cannot capture all types of barriers, particularly those requiring human judgment, contextual understanding, or interaction-based assessment. As a result, the findings should be interpreted as indicators of accessibility barriers rather than a complete evaluation of user experience.

The study adopts a structured, taxonomy-driven analytical approach to support cross-domain comparison. However, the interpretation of findings through literature-informed personas represents an analytical abstraction rather than direct observation of real user behavior. Persona-based interpretations are therefore used to support comparative reasoning about potential user impact, not to make definitive claims about lived user experience.

The custom analytical platform developed in the study is used to support data collection, normalization, and analysis. It is not evaluated as an independent artifact, and the study does not aim to assess its effectiveness as a standalone tool. Its role is methodological, enabling consistency, comparability, and reproducibility across evaluation runs.

Finally, the analysis is bounded by the selected domains, sampling strategy, and evaluation configuration. Differences observed between domains should therefore be interpreted within the context of the study design rather than generalized to all public-sector digital services.

1.5. Thesis Organization

This thesis is organized into seven chapters, each addressing a specific area of the research process.

Chapter 1 – Introduction - It presents the background, problem formulation, and motivation for the study. It outlines the research purpose, research questions, scope, and limitations of the thesis.

Chapter 2 – Literature Review - It gives an overview of existing research on web accessibility, public-sector digital services, WCAG standards, automated accessibility evaluation, and persona-based accessibility interpretation. It also explains key gaps in prior research that this thesis aims to fill.

Chapter 3 – Theoretical Framework - It introduces the conceptual foundations of the study. It presents the taxonomy framework, persona framework, domain comparison framework, and the overall analytical model used to structure and interpret accessibility barriers.

Chapter 4 – Methodology - It describes the research design, data collection process, sampling strategy, evaluation configuration, and analytical procedures. It also discusses reliability, validity, and ethical considerations.

Chapter 5 – Empirical Findings - It presents the results of the analysis. It reports findings related to the extent of accessibility barriers, the distribution of barrier types, persona-based interpretations, and cross-domain comparisons.

Chapter 6 – Discussion - It interprets the findings in relation to existing literature and the research questions. It also reflects on methodological aspects and outlines practical implications and recommendations born out of the study.

Chapter 7 – Conclusion summarizes the key findings, highlights the contributions of the study, and future research.

2. Literature review

2.1. Web Accessibility and Public Sector Context

Digitalization has made web-based public services central to social participation, service access, and administrative interaction. Public institutions, including municipalities, universities, and healthcare organizations, increasingly depend on websites as their primary information and service interface (Mergel, Edelmann, & Haug, 2019). In this context, accessibility is not a peripheral quality feature; it affects whether citizens can practically use essential public services.

The accessibility literature links this issue to both technical design and social inclusion. Accessibility is defined as the extent to which digital content can be perceived, understood, navigated, and operated by users with diverse abilities (W3C, 2024). At the same time, studies on digital inequality emphasize that exclusion is shaped not only by disability, but also by broader capability differences, literacy gaps, and uneven access to digital resources (Van Deursen & Helsper, 2015; Lazar, Goldstein, & Taylor, 2015). This dual perspective is important: inaccessible interfaces are technical artifacts, but their consequences are social and institutional (Lazar, Goldstein, & Taylor, 2015; Fernández-Díaz et al., 2023).

For public institutions, this creates a socio-technical responsibility. Accessibility concerns legal compliance and technical quality, but also democratic legitimacy in digital service provision. This thesis therefore treats public-sector web accessibility as an Information Systems problem where technical implementation, organizational practice, and user inclusion are analytically intertwined.

2.2. WCAG 2.2 and Accessibility Compliance

WCAG remains the dominant normative framework for web accessibility evaluation. Its four principles which are perceivable, operable, understandable, and robust (often referred to as the POUR principles), and three conformance levels (A, AA, AAA) provide the most widely accepted structure for assessing digital accessibility (W3C, 2024). The principles define that content must be perceivable by users (e.g., through alternatives to visual or auditory content), operable through different input methods (e.g., keyboard accessibility), understandable in terms of content and interaction, and robust enough to be reliably interpreted by a wide range of user agents, including assistive technologies. The conformance levels represent increasing degrees of accessibility requirements, with level AA commonly treated as the practical benchmark in public-sector settings.

Regulatory adoption has strengthened WCAG's institutional relevance. In the European context, the Web Accessibility Directive and related national implementations, including Swedish legislation, have formalized accessibility obligations for public services (Jonsson, Gustavsson, & Gulliksen, 2023). However, the literature shows a persistent

implementation gap: legal and standards alignment does not automatically produce accessible user experiences.

Empirical studies consistently report recurring WCAG-related violations, especially around text alternatives, contrast, semantic markup, and navigation structure (Campoverde-Molina et al., 2023; Inal et al., 2022; Kesswani & Kumar, 2022). A key tension in this literature is that WCAG functions well as a normative architecture but less well as a standalone explanation of practical accessibility outcomes. As systems become more dynamic and interaction-heavy, interpretation complexity increases, and some criteria require human judgment beyond automated checking (Fischer et al., 2025). Thus, compliance frameworks are necessary but not sufficient for understanding accessibility barriers in use.

2.3. Automated Accessibility Evaluation in Prior Studies

Automated tools such as Axe (Deque Systems, n.d.) , WAVE (WebAIM, n.d.), and Lighthouse (Google, n.d.) are widely used because they enable programmatic accessibility evaluation across many web pages, typically through APIs or automated scripts (Campoverde-Molina et al., 2023; Macakoğlu et al., 2022). For comparative studies, this scalability is a major advantage, as it supports systematic and consistent data collection across multiple pages and institutions (Inal et al., 2022).

Yet the literature also identifies major constraints in automated accessibility evaluation, including partial WCAG coverage, limited contextual interpretation, and variation between evaluation engines (Vigo, Brown, & Conway, 2013; Campoverde-Molina et al., 2023; Abu Doush et al., 2023; Fischer et al., 2025). First, automated tools cover only part of WCAG and cannot reliably assess criteria requiring contextual or semantic judgment (Fischer et al., 2025). For example, tools can detect the presence of alternative text for images, but cannot determine whether the text is meaningful or appropriate in context. Similarly, they may identify form labels or headings, but cannot evaluate whether instructions are clear, content is understandable, or navigation is logically structured from a user perspective. Second, tool outputs are not fully interchangeable; different engines produce different findings for the same page because of varying detection logic, rule coverage, and implementation approaches (Campoverde-Molina et al., 2023; Inal et al., 2022; Kumar, Shree, & Biswas, 2021). Third, reporting often remains at the level of raw issue lists, which limits interpretability for organizational prioritization.

These limitations create a methodological dilemma. Automation is necessary for scale and reproducibility, but insufficient for complete accessibility inference. Prior research therefore recommends combining automated and manual approaches where possible (Campoverde-Molina et al., 2023; Fischer et al., 2025). In practice, however, resource constraints frequently push organizations and studies toward primarily automated

evaluation. This increases the importance of analytical designs that make automated outputs more interpretable, rather than treating them as final evidence in isolation.

2.4. Taxonomy and Structured Accessibility Classification

Automated accessibility evaluation produces large volumes of heterogeneous and tool-specific findings. Different evaluation tools operationalize WCAG criteria through varying detection logic, rule definitions, and reporting structures, resulting in outputs that are not always directly comparable (Campoverde-Molina et al., 2023; Kumar, Shree, & Biswas, 2021). As a result, accessibility findings are often presented as fragmented technical issue lists, limiting interpretability and making systematic comparison across studies, domains, and institutions more difficult.

Within Information Systems research, taxonomies and classification frameworks are commonly used to organize complex phenomena into structured analytical constructs that support abstraction, comparability, and interpretation (Nickerson, Varshney, & Muntermann, 2013). In accessibility research, structured categorization approaches can help organize raw accessibility findings into broader issue types and analytical groupings, supporting clearer interpretation of recurring accessibility barriers across websites and domains (Campoverde-Molina et al., 2023; Inal et al., 2022).

Such structuring approaches are particularly relevant in large-scale accessibility evaluation, where raw outputs from automated tools may otherwise remain difficult to interpret beyond isolated technical detections. Organizing findings into broader categories can support comparative analysis, identification of recurring barrier patterns, and prioritization of accessibility improvement efforts (Campoverde-Molina et al., 2023; Inal et al., 2022).

Taken together, prior research indicates a need for more structured approaches to organizing and interpreting accessibility findings across domains and evaluation contexts. In response to this gap, the present study adopts a taxonomy-driven perspective to support systematic comparison and interpretation of accessibility barriers.

2.5. Accessibility Barriers in Public Sector Websites

A substantial body of work shows that accessibility barriers persist across public-sector web ecosystems, including municipal, educational, and healthcare contexts (Campoverde-Molina et al., 2023; Inal et al., 2022; Valtolina & Fratus, 2022). The literature is therefore consistent on prevalence, but less consistent on comparative explanation.

Most studies categorize barriers through WCAG's principle structure. Frequently reported patterns include perceivability problems (for example missing alternative text and contrast issues), operability constraints (for example keyboard and navigation problems), understandability challenges (for example complex language and unclear instructions),

and robustness failures in semantic or assistive-technology compatibility (Akgül, 2021; Alajarmeh, 2021; Ismail & Kuppusamy, 2022; Máñez-Carvajal et al., 2021; Campoverde-Molina et al., 2023; Laamanen et al., 2024).

However, prior studies often remain domain-specific or single-domain descriptive, which limits systematic cross-domain comparison. Claims about domain-specific causes (for example resource constraints, structural complexity, or service criticality) are plausible but unevenly operationalized across studies (Fakrudeen, 2024; Jonsson et al., 2023). This indicates a need for analytical approaches that can compare domains under a common model while preserving barrier-level detail.

2.6. Persona-Informed Accessibility Research

Persona-based approaches are used to interpret accessibility findings in user-relevant terms. Personas are typically defined as evidence-informed user archetypes that represent recurring needs, constraints, and interaction patterns (Pruitt & Adlin, 2006; Miaskiewicz & Kozar, 2011). In accessibility work, they can help translate technical violations into likely impact profiles across user groups. This interpretive function is important because accessibility barriers are not uniformly experienced. For example, visual, cognitive, and assistive-technology-related challenges are linked to different barrier types and interaction consequences (Henry, Abou-Zahra, & Brewer, 2014; Lazar et al., 2015). Personas can therefore improve analytical relevance when direct user testing is not feasible (Marshall et al., 2015).

At the same time, persona literature warns against overreach. Persona models can oversimplify heterogeneous user realities or reinforce stereotypes if weakly grounded (Turner & Turner, 2011). For accessibility research, this implies a methodological boundary: personas should be treated as structured interpretive lenses, not as substitutes for direct user evidence. This thesis adopts that bounded use.

This positions personas as a structured mechanism for linking technical accessibility findings to user-relevant impact, forming a conceptual basis for their use in subsequent analysis.

2.7. Research Gap and Chapter Summary

Existing research establishes that accessibility barriers in public-sector websites are persistent and that automated methods are operationally necessary for large-scale analysis. It also shows clear limitations in current practice: partial criterion coverage, inconsistent tool outputs, weak cross-domain comparability, and reporting styles that often stop at technical issue inventories (Fischer et al., 2025; Campoverde-Molina et al., 2023; Inal et al., 2022; Valtolina & Fratus, 2022).

The key gap is therefore not the absence of accessibility studies, but the absence of an integrated analytical approach that simultaneously provides (1) scalable empirical comparison across domains, (2) structured normalization of findings into interpretable barrier patterns, and (3) user-impact-oriented interpretation under explicit methodological boundaries.

This thesis addresses that gap through a barrier-oriented, taxonomy-driven, cross-domain design with persona-informed interpretation. It does not position automated findings as complete user evidence; rather, it uses them as structured barrier signals that can be compared and interpreted more meaningfully across Swedish public-service contexts.

3. Theoretical framework

3.1. Conceptual Foundations

This thesis is grounded in a socio-technical view of web accessibility. Accessibility is treated neither as a purely technical checklist issue nor as a purely subjective experience issue, but as an interaction between technical design decisions, institutional web practices, and diverse user capabilities (Mergel, Edelmann, & Haug, 2019; Lazar, Goldstein, & Taylor, 2015). In this view, technical violations are analytically meaningful because they indicate potential barriers, while user-centered interpretation is needed to understand how those barriers may affect real interaction and participation (Fernández-Díaz et al., 2023).

The framework therefore combines three layers:

1. A standards layer based on WCAG as a normative structure for accessibility quality
2. An empirical layer based on observed automated findings across websites
3. An interpretive layer based on personas and impact groups to connect technical barriers to likely user consequences.

This layered perspective supports the thesis objective of producing findings that are both technically grounded and analytically interpretable across institutional domains (W3C, 2024; Van Deursen & Helsper, 2015).

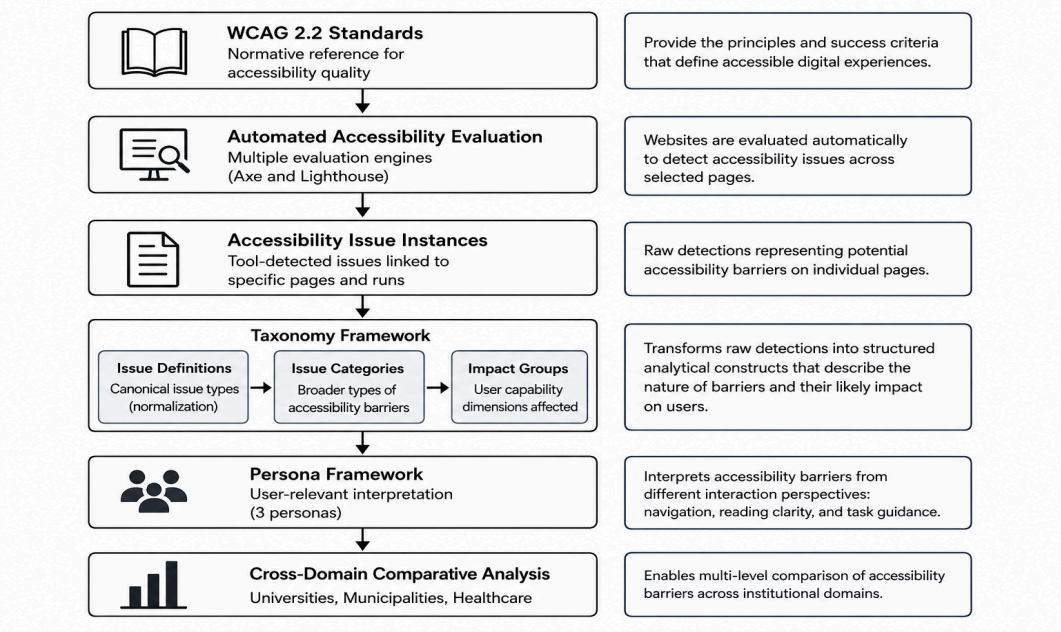


Figure 3.1. Conceptual structure of the theoretical and analytical framework.

3.2. Accessibility As Barrier-Oriented Analysis

The analytical focus of the thesis is barrier-oriented rather than compliance-only. Compliance status alone can indicate whether standards are met, but it does not sufficiently explain which barriers dominate, how barriers differ by domain, or which user groups are more likely to be affected by recurring patterns (Campoverde-Molina et al., 2023; Inal et al., 2022).

Accordingly, each detected issue is treated as a barrier signal. Barrier analysis in this thesis proceeds through:

- Identification of recurring issue types
- Classification into broader issue categories
- Mapping to impact groups
- Interpretation through persona lenses

This approach allows the study to answer not only how much non-compliance exists, but also what kind of accessibility burden is present and which user groups are likely to be most affected (Fischer et al., 2025).

3.3. Taxonomy Framework

To enable systematic cross-domain comparison, this study employs a structured taxonomy that standardizes how automated accessibility findings are represented and analyzed. The taxonomy addresses a key limitation in prior research, where accessibility issues are often reported as tool-specific and unstructured outputs, limiting comparability and interpretability (Campoverde-Molina et al., 2023; Valtolina & Fratus, 2022).

Rather than treating raw tool detections as final analytical units, the taxonomy provides a consistent representation of accessibility findings that supports structured aggregation and comparison across websites and domains. This enables accessibility barriers to be analyzed at multiple levels of granularity within a unified analytical framework.

The taxonomy was developed through an iterative and empirically informed process during the early stages of accessibility evaluation. Initial exploratory evaluations were used to identify recurring accessibility findings and overlaps between tool-specific outputs. Based on these findings, issue definitions were inductively grouped into broader analytical categories representing recurring types of accessibility barriers. The taxonomy was refined iteratively as additional evaluations were conducted, with categories and mappings adjusted to improve consistency, reduce conceptual overlap, and maintain alignment with WCAG principles and success criteria.

3.3.1. Taxonomy Structure

The taxonomy consists of four primary analytical units:

- **Issue instances**:
  Individual detections produced by automated evaluation tools, tied to specific pages and evaluation runs. These represent the empirical observations of accessibility barriers.
- **Issue definitions**:
  Canonical representations of issue types used to normalize heterogeneous tool outputs. Multiple tool-specific detections may map to the same issue definition, ensuring consistency across evaluation engines.
- **Issue categories**:
  Higher-level groupings of issue definitions that represent broader types of accessibility barriers (e.g., structure, navigation, forms). These categories describe the *nature* of the barrier.
- **Impact groups**:
  Analytical groupings representing user capability dimensions affected by accessibility barriers (e.g., visual, cognitive, motor, auditory). These groups describe the *potential consequences* of barriers for different users.

Together, these elements form a multi-dimensional taxonomy that separates technical detection, barrier type, and user-impact perspective into distinct but linked analytical layers.

The operationalization of issue categories and their relation to WCAG principles and success criteria is presented in Table 3.1.

Table 3.1. Accessibility issue categories and their relation to WCAG 2.2

Category	Description	WCAG Principles	WCAG Success Criteria
Forms and input	Issues affecting form controls, labels, input purpose, and task completion	Understandable, Operable	3.3.2, 3.3.7, 2.1.1
Errors and feedback	Issues related to error identification, messaging, and recovery guidance	Understandable	3.3.1, 3.3.3, 3.3.4
Keyboard and interaction	Issues affecting keyboard access, focus management, and interaction behavior	Operable	2.1.1, 2.1.2, 2.4.7
Navigation, orientation and consistency	Issues affecting navigation structure, orientation, and predictable behavior	Operable, Understandable	2.4.1, 2.4.6, 3.2.3

Structure and semantics	Issues related to headings, landmarks, and semantic organization	Perceivable, Robust	1.3.1, 1.3.2
Names and text alternatives	Issues involving accessible names and text alternatives for non-text content	Perceivable, Robust	1.1.1, 4.1.2
Media and multimedia	Issues concerning audio, video, captions, and transcripts	Perceivable	1.2.1, 1.2.2, 1.2.5
Visual design and contrast	Issues affecting contrast, readability, and visual perception	Perceivable	1.4.3, 1.4.10, 1.4.11
Language and comprehension	Issues involving language declaration and cognitive readability	Understandable	3.1.1, 3.1.5
Motion, animation and timing	Issues related to time limits, animations, and motion sensitivity	Operable	2.2.1, 2.3.1, 2.3.3
Compatibility and robustness	Issues affecting parsing and assistive technology compatibility	Robust	4.1.1, 4.1.2, 4.1.3

The impact groups used in this study are defined in Table 3.2.

Table 3.2. Accessibility impact groups

Impact groups	Description
Visual	Impact on visual perception, including low vision and reliance on visual presentation
Cognitive	Impact on understanding, memory, attention, and task completion clarity
Motor	Impact on physical interaction, including keyboard use and pointer control
Auditory	Impact on hearing-related access to audio content and media

3.3.2. Mapping Logic

Each issue definition is systematically mapped to one issue category and one or more impact groups. This mapping is defined through a structured configuration process based on WCAG principles, accessibility literature, and the functional characteristics of each issue type.

The decision to assign each issue definition to a single category is motivated by the need for analytical clarity and mutual exclusivity. Allowing multiple category assignments

would introduce overlap and reduce interpretability in aggregation and comparison. In contrast, mapping to multiple impact groups is permitted, as accessibility barriers often affect multiple user capabilities simultaneously.

This asymmetry reflects a deliberate distinction: issue categories describe the type of accessibility barrier, whereas impact groups represent the user capabilities most likely to be affected.

3.3.3. Relationship to WCAG

Where applicable, issue definitions are linked to relevant WCAG success criteria. However, the taxonomy does not strictly replicate the WCAG structure. Instead, it operates at the level of tool-detected issues, which are often more granular or operationalized differently than formal WCAG criteria.

This design choice reflects the reality that automated tools interpret WCAG through implementation-specific checks that do not always map one-to-one to individual success criteria. Retaining issue definitions beyond strict WCAG mappings ensures that relevant accessibility signals are preserved, supporting the study's barrier-oriented analytical approach.

3.3.4. Theoretical Role of the Taxonomy

The taxonomy serves two primary theoretical functions:

- **Normalization of heterogeneous data:** By mapping tool-specific outputs to shared issue definitions, the taxonomy reduces variability between evaluation engines and creates a consistent analytical foundation.
- **Abstraction and interpretability:** By linking issue definitions to categories and impact groups, the taxonomy enables transformation from low-level technical detections to higher-level constructs that can be meaningfully interpreted and compared across domains.

This layered abstraction is essential for addressing the limitations of automated accessibility evaluation, where raw outputs are often fragmented and difficult to interpret in isolation (Inal et al., 2022; Campoverde-Molina et al., 2023).

3.3.5. Design Considerations and Boundaries

The taxonomy is intentionally designed to balance completeness, interpretability, and analytical consistency. It prioritizes structured comparability over an exhaustive representation of all possible accessibility nuances, ensuring that findings can be consistently analyzed across domains. At the same time, the taxonomy introduces a level of researcher-defined abstraction, particularly in the mapping of issue definitions to categories and impact groups. While this abstraction enables systematic analysis, it also reflects an intentional simplification of complex accessibility phenomena. Accordingly, the taxonomy does not aim to capture the full complexity of accessibility experiences, but

rather to provide a consistent analytical model for interpreting detectable barrier patterns. These design choices support the study's objective of enabling systematic cross-domain comparison while preserving interpretability and methodological transparency.

3.3.6. Summary

The taxonomy framework establishes a structured representation of accessibility findings that enables consistent transformation of raw automated detections into interpretable analytical constructs. By separating issue detection, barrier type, and user-impact dimensions, it supports systematic aggregation and comparison of accessibility barriers across public-sector domains. As such, the taxonomy forms a foundational component of the study's analytical model and underpins subsequent interpretation and cross-domain analysis.

3.4. Persona Framework

To complement the technical analysis of accessibility barriers, this study adopts a persona-informed interpretive framework that links normalized issue patterns to user-relevant perspectives. The purpose of this framework is not to represent individual users or empirical user groups, but to provide a structured mechanism for interpreting how different types of accessibility barriers may affect distinct interaction contexts.

Personas are widely used in design and Information Systems research as evidence-informed abstractions of user needs, constraints, and interaction patterns (Pruitt & Adlin, 2006; Miaskiewicz & Kozar, 2011). In accessibility research, they can support the translation of technical findings into user-relevant interpretations by highlighting how specific barrier types relate to different forms of interaction difficulty (Henry, Abou-Zahra, & Brewer, 2014; Lazar et al., 2015).

However, persona-based approaches also introduce methodological risks. Prior research has shown that personas may oversimplify heterogeneous user realities or imply unwarranted assumptions about user behavior if not carefully bound (Turner & Turner, 2011). To address this limitation, personas in this study are explicitly defined as analytical constructs, rather than representations of real users or substitutes for empirical user data.

3.4.1. Operationalization of Personas

Within this study, personas are operationalized as structured configurations of relevance across issue categories and impact groups. Each persona is defined by a subset of issue categories and associated impact groups that represent a particular interaction perspective.

In practical terms, persona interpretation is implemented as a rule-based mapping process applied to issue definitions. An issue is considered relevant for a given persona if it belongs to one of the persona's defined categories and is associated with one or more of its corresponding impact groups. This enables consistent identification and aggregation of persona-relevant accessibility barriers across websites and domains.

This operationalization ensures that persona interpretation remains systematic through predefined mapping rules, reproducible through consistent application across the dataset, and transparent through the explicit configuration of persona definitions.

3.4.2. Persona Definitions

The study employs three personas, selected to represent distinct dimensions of accessibility-related interaction challenges identified in prior literature. The Assistive Technology Navigation Persona focuses on barriers affecting navigation and interaction through assistive technologies and non-pointer input methods. The Reading Clarity Persona emphasizes issues related to content readability, visual clarity, and comprehension of written information. The Task Guidance Persona highlights barriers affecting task-oriented interaction, including form completion, input handling, and error recovery.

These personas are not intended to capture the full diversity of user experiences. Instead, they provide analytically distinct perspectives that reflect recurring accessibility concerns related to navigation, comprehension, and task execution.

The operational configuration of personas in terms of issue categories and impact groups is presented in Table 3.3.

Table 3.3. Operationalized personas based on issue categories and impact groups

Persona	Associated Issue Categories	Associated impact groups	Interpretive focus	Literature basis
Assistive Technology Navigation Persona	Navigation, orientation and consistency; Keyboard and interaction; Structure and semantics; Names and text alternatives; Compatibility and robustness	Visual; Motor	Ability to navigate and interact with interfaces using assistive technologies and non-pointer input methods	Informed by literature discussing assistive-technology interaction, semantic navigation, and accessibility requirements for screen-reader and keyboard-based interaction (Henry, Abou-Zahra, & Brewer, 2014; Lazar, Goldstein, & Taylor, 2015; W3C, 2024)
Reading Clarity Persona	Language and comprehension; Visual design and contrast;	Visual; Cognitive	Readability, clarity of content, and ease of understanding	Informed by literature discussing readability, understandability, cognitive

	Structure and semantics		written information	accessibility, and differences in users' ability to access and benefit from digital information (Van Deursen & Helsper, 2015; Henry, Abou-Zahra, & Brewer, 2014; Lazar, Goldstein, & Taylor, 2015)
Task Guidance Persona	Forms and input; Errors and feedback; Navigation, orientation and consistency	Cognitive; Motor	Clarity of task flows, guidance, and ability to complete processes without confusion or error	Informed by literature discussing accessible interaction design, task completion barriers, input guidance, and usability-related accessibility challenges in digital services (Henry, Abou-Zahra, & Brewer, 2014; Lazar, Goldstein, & Taylor, 2015)

3.4.2.1. Assistive Technology Navigation Persona

This persona represents users who rely on assistive technologies, such as screen readers, and non-pointer input methods when navigating and interacting with public-sector websites.

Table 3.4. Assistive Technology Navigation Persona

Attribute	Description
Representative user context	User relying on assistive technologies such as screen readers and keyboard navigation to access and navigate public-sector websites
Primary goals	Navigate websites efficiently and independently using non-pointer input methods
Accessibility needs	Semantic page structure, logical heading hierarchy, meaningful landmarks and link descriptions, keyboard-accessible interaction, and compatibility with assistive technologies

Typical barriers	Missing or duplicated landmarks, incorrect heading structure, poor semantic organization, unclear navigation structure, and compatibility issues affecting assistive technologies

3.4.2.2 Reading Clarity Persona

This persona represents users affected by barriers related to readability, visual clarity, and comprehension when accessing information on public-sector websites.

Table 3.5. Reading Clarity Persona

Attribute	Description
Representative user context	User affected by barriers related to readability, visual clarity, and comprehension of digital content
Primary goals	Read and understand public-sector information clearly and efficiently
Accessibility needs	Sufficient color contrast, readable text presentation, predictable content structure, and understandable language
Typical barriers	Low color contrast, poor heading organization, visually unclear layouts, inconsistent content structure, and language-related comprehension issues

3.4.2.3. Task Guidance Persona

This persona represents users experiencing difficulties completing digital tasks because of unclear forms, insufficient guidance, or inaccessible interaction flows on public-sector websites.

Table 3.6. Task Guidance Persona

Attribute	Description
Representative user context	User experiencing difficulties completing digital tasks because of unclear forms, insufficient guidance, or inaccessible interaction flows
Primary goals	Complete digital processes and service-related tasks accurately and without confusion
Accessibility needs	Clearly labeled forms, understandable instructions, meaningful error feedback, and predictable interaction flows
Typical barriers	Missing form labels, unclear instructions, inadequate error guidance, inaccessible interactive elements, and confusing task flows

3.4.3. Role in the Analytical Model

The persona framework serves two primary functions within the analytical model.

First, it provides an interpretive layer that translates technical accessibility findings into user-relevant perspectives. By linking issue patterns to defined interaction contexts, the framework supports a more meaningful understanding of how accessibility barriers may affect different types of users.

Second, it enables comparative analysis across domains by allowing accessibility burden to be examined from multiple interaction perspectives. This makes it possible to identify whether certain domains disproportionately affect specific types of interaction, such as navigation, comprehension, or task completion.

Importantly, persona-based findings are interpreted as analytical approximations of user impact, rather than direct evidence of user experience. The framework does not replace user testing or empirical observation, but provides a structured method for interpreting large-scale accessibility data in the absence of such data.

3.4.4. Summary

The persona framework extends the taxonomy-driven analysis by introducing a structured interpretive layer that connects technical accessibility findings to user-relevant perspectives. By operationalizing personas as rule-based mappings over issue categories and impact groups, the framework enables consistent, transparent, and reproducible interpretation of accessibility barriers across domains. This supports the study's objective of moving beyond technical issue reporting toward a more user-oriented understanding of accessibility in public-sector digital services.

3.5. Domain Comparison Framework

Domain comparison is central to the thesis design. Universities, municipalities, and healthcare organizations are treated as institutionally distinct but comparable digital service contexts.

The framework conceptualizes domain differences as emerging through variations in content and service complexity, interaction flows, governance practices, and technical platform choices. These factors are not directly measured in this study, but are treated as contextual conditions that may contribute to differences in accessibility patterns across domains.

Comparative analysis is performed using the structured taxonomy and persona framework introduced in previous sections. This enables comparison across domains at multiple analytical levels:

- Overall burden level: issue prevalence and distribution across evaluated pages

- Category distribution: the relative dominance of different types of accessibility barriers
- Impact-group profile: the distribution of issues across user capability dimensions
- Persona-linked burden patterns: the extent to which specific interaction perspectives are disproportionately affected

By linking domain comparison to issue categories, impact groups, and persona configurations, the framework supports a consistent analytical approach in which differences between domains are interpreted as variations in barrier structure rather than simple differences in compliance rates.

The domain characteristics outlined above provide a theoretical basis for interpreting variation in accessibility profiles, without being used as explicit analytical variables. This allows the analysis to remain grounded in observable data while still offering theoretically informed explanations.

This multi-level design allows the thesis to move beyond simplistic ranking logic and instead describe structural differences in accessibility profiles across domains (Inal et al., 2022; Jonsson, Gustavsson, & Gulliksen, 2023).

3.6. Analytical Model

The analytical model defines the structured transformation process through which raw accessibility evaluation data is converted into interpretable, comparable insights. It integrates automated data collection with taxonomy-driven normalization and persona-informed interpretation, forming a multi-stage analytical pipeline aligned with the study's research objectives.

The model is designed to address known limitations of automated accessibility evaluation, particularly partial coverage and limited interpretability of raw issue outputs (Campoverde-Molina et al., 2023; Fischer et al., 2025). Rather than treating automated findings as final results, the model interprets them as structured barrier indicators that require transformation and aggregation to support meaningful analysis.

3.6.1. Analytical Pipeline

The model consists of five sequential stages

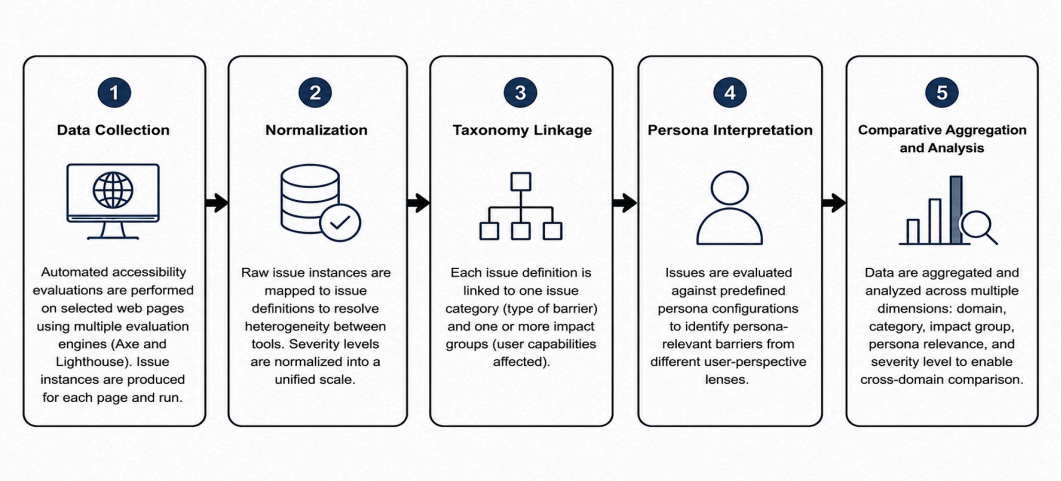


Figure 3.2. Analytical pipeline for transforming accessibility findings into structured comparative analysis.

1. Data collection
Automated accessibility evaluations are performed on selected web pages using multiple evaluation engines. Each evaluation produces *issue instances*, representing detected accessibility violations linked to specific pages and runs. These instances form the raw empirical dataset.

2. Normalization
Raw issue instances are mapped to *issue definitions*, which represent canonicalized issue types independent of specific tool outputs. This step resolves heterogeneity between evaluation tools by consolidating equivalent or related detections into shared analytical units. Severity levels are also normalized into a unified scale to enable consistent comparison across tools.

3. Taxonomy linkage
Each issue definition is systematically linked to:

- a single issue category, representing the type of accessibility barrier
- one or more impact groups, representing the user capabilities likely to be affected.

This mapping transforms low-level technical detections into structured analytical constructs, enabling aggregation and comparison across websites and domains.

4. Persona interpretation
Issue definitions are evaluated against predefined *persona configurations*, which specify relevant combinations of issue categories and impact groups. This rule-based mapping

identifies which issues are relevant for each persona, enabling aggregation of accessibility burden from different user-perspective lenses.

Persona interpretation is used as an analytical abstraction to support structured inference about potential user impact. It does not represent direct measurement of user experience, but provides a consistent mechanism for interpreting accessibility patterns across domains.

5. Comparative aggregation and analysis
Normalized and interpreted data are aggregated across multiple analytical dimensions, including:

- domain (universities, municipalities, healthcare)
- issue category
- impact group
- persona relevance
- severity level

This enables multi-level comparison of accessibility barriers, supporting both descriptive profiling and cross-domain analysis aligned with the research questions.

3.6.2. Units of Analysis

The model operates across multiple analytical levels, where each level serves a distinct analytical purpose. At the issue-instance level, the analysis focuses on the frequency and intensity of detected accessibility barriers. This level captures how often issues occur and the extent to which particular barriers are represented across evaluated pages.

At the issue-definition level, the focus shifts from individual occurrences to the diversity and typology of accessibility barriers. This enables comparison of the different kinds of issues identified by the evaluation tools rather than only their frequency.

The category and impact-group level supports higher-level structural comparison by grouping related issues into broader analytical constructs. This makes it possible to compare how accessibility barriers are distributed across different domains and categories of impact.

Finally, the persona level enables interpretation of accessibility burden from different user perspectives by relating findings to the types of users potentially affected by the identified barriers.

Maintaining these distinct units of analysis prevents analytical conflation and ensures that comparisons are made between comparable constructs across the different levels of the model.

3.6.3. Model Boundaries and Assumptions

The analytical model includes explicit boundaries to support valid interpretation of the findings. Automated findings are treated as indicators of accessibility barriers rather than complete representations of user experience. Similarly, persona mappings are interpreted as analytical approximations of potential user impact, not as empirical evidence of actual user behavior or experience. In addition, comparative findings are bound to the sampled websites and the specific evaluation configuration used in the study.

These constraints acknowledge the partial nature of automated accessibility evaluation while preserving the model's value for structured and comparative analysis.

3.7. Summary

The analytical model provides a reproducible framework for transforming heterogeneous accessibility evaluation outputs into structured, interpretable insights. By linking technical detections to taxonomy-based constructs and persona-informed interpretations, the model supports systematic comparison of accessibility barriers across public-sector domains. This enables the study to move beyond compliance-oriented reporting toward a more comprehensive understanding of accessibility as a structured and domain-dependent phenomenon.

4. Methodology

4.1. Philosophical Tradition

From a research-philosophy perspective, the study is positioned in a quantitative post-positivist tradition, where accessibility issues are treated as observable and measurable barrier signals that can be systematically compared across cases (Bryman & Bell, 2015; Creswell & Creswell, 2018). This stance assumes that automated findings provide empirically useful indicators, while still acknowledging that any measurement model is partial and interpretation-dependent. In this design, personas are used as analytical lenses for structured interpretation, not as subjective user data or direct substitutes for participant-based evidence.

4.2. Methodological Approach

This study followed a quantitative comparative research design focused on accessibility barriers in Swedish public institution websites. A quantitative comparative research design involves the systematic collection and analysis of numerical data across multiple cases in order to identify patterns, similarities, and differences (Bryman & Bell, 2015; Creswell & Creswell, 2018). In this study, the design supports systematic comparison across institutional domains by analyzing structured accessibility findings collected under a standardized evaluation setup (Inal et al., 2022; Valtolina & Fratus, 2022).

To support consistent data collection and comparative analysis across multiple accessibility evaluation tools, a software platform was developed as part of the project. The software platform developed in the project was used as a research instrument, not as an independent artifact under evaluation. Its role in the methodology was to ensure consistency in data collection, storage, normalization, and export. The methodological objective was not only to measure accessibility findings, but also to make those findings analytically comparable and reproducible (Campoverde-Molina et al., 2023; Fischer et al., 2025).

Accordingly, this study is not framed as Design Science Research. The platform is not evaluated as the primary contribution; it is the means through which empirical comparative analysis is conducted under controlled and reproducible conditions.

4.3. Methods for Data Collection

4.3.1. Study Context And Sampling Strategy

The empirical scope covered three domains of Swedish public service websites: universities, municipalities, and healthcare organizations. To support balanced cross-domain analysis, the sampling strategy was stratified by domain with equal allocation. The final sample included 36 websites in total, distributed as 12 websites per domain. The full list of evaluated websites is provided in Appendix D (Table D1).

For each sampled website, the intended page scope was 30 pages. The 30-page threshold was chosen as a methodological balance between breadth and feasibility. It was intended to capture sufficient variation in page structures and content types within each website while keeping the evaluation process computationally manageable and comparable across websites. Across 12 sampled websites per domain, this resulted in approximately 360 evaluated pages per domain and more than 1,000 pages overall, providing a sufficiently broad dataset for cross-domain comparative analysis. Page selection followed a structured two-step procedure. First, the website's homepage URL was manually specified as the seed URL, serving as the entry point for the evaluation. Second, additional pages were identified through automated discovery based on the website's sitemap.

The discovery process was constrained to ensure consistency and relevance. Only pages within the same domain as the seed URL were included, and pages outside the primary language scope were excluded. If the seed URL was located within a specific path, only pages within that path were considered.

Discovered URLs were filtered to exclude non-relevant or system-generated content, such as login pages, administrative paths, search results, and non-HTML resources. Only publicly accessible HTML pages returning valid responses were included in the final sample.

To improve representativeness, pages were selected to capture a range of content types and site structures, rather than concentrating on similar pages within a single section. The process continued until the target number of pages was reached, or until no additional eligible pages were available within the defined scope.

This approach balanced control and scalability. It reduced the risk of arbitrary manual page picking while preserving methodological consistency across all websites. When a website could not provide the full intended page scope because of sitemap structure, redirects, or other technical constraints, this was treated as a coverage characteristic and documented as part of methodological transparency. This is consistent with prior accessibility-evaluation work that emphasizes transparent handling of coverage variation in large-scale automated studies (Inal et al., 2022; Campoverde-Molina et al., 2023).

4.3.2. Data Collection Pipeline

Data collection was conducted through structured evaluation runs for each configured website. Within each run, the selected page scope was scanned using two automated accessibility engines, Axe and Lighthouse. The combined use of these engines was intended to widen detection coverage while keeping collection procedures uniform across all domains, reflecting known tool-coverage differences in the literature (Campoverde-Molina et al., 2023; Inal et al., 2022).

Findings were persisted with run-level and page-level context, including coverage outcomes and normalized issue records. Coverage information was explicitly retained in order to support quality-aware interpretation of results. This was important because low issue counts can be misleading when scan success is low. By preserving coverage metadata, the analysis could distinguish between genuinely lower barrier burden and incomplete page evaluation, an issue frequently noted in automated accessibility assessment research (Fischer et al., 2025).

4.3.3. Evaluation Configuration

The same core evaluation configuration was applied throughout the study to preserve internal comparability. In concrete terms, websites were evaluated under a shared run logic, shared discovery principles, and shared page-target strategy. This consistency was necessary to ensure that observed differences between domains could be interpreted as empirical differences rather than configuration artifacts (Valtolina & Fratus, 2022; Inal et al., 2022).

Discovery behavior was designed to support realistic but controlled page sampling. The heuristic considered sitemap sources, normalized candidate URLs, applied language-scope matching from the seed URL as language variations can affect both the semantic meaning and structural organization of web pages, including navigation, content grouping, and localized layouts, filtered low-value/system paths, and prioritized structural diversity in selected pages. The purpose of these controls was methodological: to produce page sets that are broad enough for cross-domain analysis while filtering out irrelevant pages such as XML, PDF or login pages, duplicate structures, or off-scope language sections. This aligns with recommendations to increase comparability and interpretability in automated, multi-site accessibility designs (Campoverde-Molina et al., 2023; Fischer et al., 2025).

4.4. Methods for Data Analysis

4.4.1. Measures And Variables

The analysis used a multi-level variable model combining technical and interpretive measures. Core variables included domain, issue burden, issue diversity, issue category, impact group, persona interpretation, severity, and WCAG alignment metadata. This variable design was selected to support both descriptive profiling and cross-domain comparison.

A key methodological decision was to separate two counting grains: issue-instance counts and unique issue-definition counts. Issue-instance counts represent burden intensity (how frequently barriers occur), whereas issue-definition counts represent typological breadth (how many distinct barrier types are present). Keeping these measures separate reduced the risk of analytical over-simplification and made comparison logic clearer.

Severity was used as an additional analytical dimension to reflect the relative impact of detected issues. Severity levels provided by the evaluation tools were treated as indicative weights of barrier importance rather than formal WCAG classifications. This enabled distinction between frequently occurring low-impact issues and less frequent but higher-impact barriers in the comparative analysis.

4.4.2. Data Processing And Normalization

Because raw outputs from different tools are not directly comparable, the methodology included a normalization layer. Detected findings were mapped to canonical issue definitions, and these definitions were linked to issue categories and impact groups. Severity levels from different tools were normalized into a unified scale to enable consistent comparison. However, linking issue definitions to issue categories and impact

groups was performed as a qualitative manual using the software platform interface, where each issue definition was reviewed and assigned to the most appropriate category and impact groups. Axe severity values were retained directly, while Lighthouse scores were mapped to severity levels using predefined thresholds: scores below 0.3 were classified as *serious*, scores below 0.7 as *moderate*, and scores of 0.7 and above as *minor*.

No critical severity level was assigned based solely on Lighthouse scores. This decision reflects the fact that Lighthouse audit scores represent aggregated quality signals rather than direct measures of user-impact severity. As such, low Lighthouse scores do not necessarily correspond to critical accessibility barriers in the same way as high-impact violations identified by Axe. Instead, critical severity levels were reserved for cases where tools explicitly report high-impact issues, ensuring a conservative and methodologically consistent estimation of barrier severity.

When multiple sources reported the same issue, the highest severity level was retained according to an ordered ranking (unknown < minor < moderate < serious < critical), ensuring a conservative estimation of barrier impact. This created a shared analytical vocabulary across tools and domains, addressing known cross-tool heterogeneity in prior work (Campoverde-Molina et al., 2023; Inal et al., 2022).

This normalization layer was implemented through structured configuration datasets defining issue definitions, issue categories, and impact groups. In addition, an explicit mapping layer was defined to link each issue definition to a corresponding issue category and one or more impact groups. This mapping enabled consistent transformation of raw issue detections into higher-level analytical constructs used for aggregation and interpretation. The complete mapping is provided in Appendix C (Table C1).

Following normalization the persona layer was then used as an interpretive lens rather than a direct behavioral measure. In other words, personas were used to structure likely user-relevance patterns in recurring barriers, not to claim direct user-experience outcomes. This use follows persona literature that supports structured interpretation while cautioning against overclaiming (Pruitt & Adlin, 2006; Turner & Turner, 2011; Marshall et al., 2015). The overall processing logic moved from issue instances to normalized definitions, then to taxonomy and persona interpretation, and finally to domain-level aggregation.

While WCAG mappings were preserved where available, issue definitions without direct WCAG references were also retained. This is because automated accessibility tools operationalize WCAG principles through more granular and heuristic checks that do not always correspond to a single success criterion. Excluding such issues would result in loss of relevant accessibility signals. Therefore, all issue definitions were included in the analysis to maintain completeness and alignment with the barrier-oriented analytical approach.

4.4.3. Analytical Procedure

The analytical procedure followed a staged comparison logic aligned with the study's research questions. The first stage established dataset context through descriptive profiling of run outcomes, page coverage, and total findings. This stage provided the empirical basis for addressing RQ1 by quantifying the overall extent of accessibility barriers across the evaluated websites.

The second stage examined domain-level barrier distributions, focusing on issue-instance counts, issue-definition diversity, and severity distribution. This supported further analysis of RQ1 by identifying differences in accessibility burden across domains.

The third stage analyzed accessibility findings through the taxonomy framework, examining the distribution of issue categories and impact groups. This stage addressed RQ2 by identifying the most prevalent types of accessibility barriers and their associated user-impact dimensions.

The fourth stage applied persona-based interpretation, mapping issue definitions to predefined persona configurations. This enabled structured interpretation of accessibility barriers from different interaction perspectives, further supporting RQ2 by linking technical findings to user-relevant impact patterns.

The fifth stage performed cross-domain comparison using category distributions, impact-group profiles, and persona-linked patterns. This stage addressed RQ3 by identifying structural differences in accessibility barrier profiles across universities, municipalities, and healthcare websites.

Finally, the results of these analytical stages were synthesized into research-question-level conclusions. This staged procedure ensured that interpretive claims remained grounded in systematically derived findings rather than inferred in a single step.

4.5. Reliability, Validity And Reproducibility

Reliability was supported by methodological consistency: the same evaluation setup across websites, balanced domain sampling, shared page-target logic, and stable normalization/interpretation rules. This reduced procedural variation and improved comparability between cases.

Construct validity was strengthened by explicitly separating technical detections from interpretive layers such as taxonomy and personas. Internal validity was supported through configuration consistency and uniform analytical procedures across domains. External validity remained bounded to the selected sample and page scope, and findings should therefore be interpreted as conclusions about the studied corpus rather than all public-sector websites (Inal et al., 2022; Valtolina & Fratus, 2022).

Reproducibility was supported through structured export artifacts that preserve datasets, variable definitions, taxonomy/persona configuration snapshots, and methods/provenance

metadata. These artifacts make it possible to reconstruct key analytical steps and independently verify descriptive outcomes.

4.6. Ethical Considerations

The study evaluated publicly accessible web pages and did not involve direct collection of personal data from participants. Ethical considerations therefore focused primarily on responsible interpretation and reporting. Automated findings were treated as barrier indicators rather than complete representations of user experience, and limitations were reported explicitly to avoid overclaiming (Fischer et al., 2025; Lazar, Goldstein, & Taylor, 2015).

An additional ethical principle was constructive framing. The analysis was oriented toward institutional learning and accessibility improvement rather than punitive ranking of organizations.

5. Empirical Findings

5.1. Overview of the Dataset

The dataset used in this analysis is summarized in Table 5.1. It consists of accessibility evaluation results from 36 Swedish public institution websites across three domains: universities, municipalities, and healthcare organizations.

A total of 4,469 accessibility issue instances were identified, representing 45 unique issue definitions. As shown in Table 5.1, accessibility issues are unevenly distributed across domains, with universities and municipalities accounting for the majority of detected issues, while healthcare websites represent a substantially smaller share.

Differences in the number of unique issue definitions indicate variation in the diversity of accessibility problems across domains. Universities exhibit the highest number of unique issues, while municipalities show lower diversity despite a similarly high total issue count. Healthcare websites, although lower in total issue instances, still exhibit a relatively broad range of issue types.

Table 5.1. Dataset overview showing the distribution of evaluated websites, accessibility issue instances, and unique issues across domains.

Domain	Websites	Pages	Issue Instances	Unique Issues per domain	Share (%)
university	12	360	1876	30	41.98
healthcare	12	360	767	27	17.16
municipality	12	360	1826	24	40.86
Total	36	1080	4469	45	100

The detailed issue matrix is provided in Appendix E (Table E1).

5.2. RQ1: Extent of Accessibility Barriers

The extent of accessibility barriers across the evaluated websites is examined through the total number of detected issue instances and their distribution across domains. A total of 4,469 accessibility issue instances were identified, indicating a substantial presence of accessibility barriers across the dataset.

5.2.1. Overall Burden

The high number of detected issue instances suggests that accessibility barriers are widespread across the evaluated public-sector websites. These findings indicate that,

despite the presence of accessibility standards and regulatory frameworks, a considerable number of detectable issues persist across digital public services.

While total issue counts provide an overall indication of accessibility burden, they must be interpreted in relation to their distribution across domains.

5.2.2. Domain Distribution

Accessibility issues are unevenly distributed across domains. Universities and municipalities exhibit similarly high levels of accessibility issues, with 1,876 and 1,826 instances respectively, accounting for the majority of detected issues. In contrast, healthcare websites show a substantially lower number of issue instances (767), representing a markedly smaller share of the total.

The distribution of issue instances is further influenced by a small number of high-frequency issue types. Across domains, issues such as *region*, *color-contrast*, and *landmark-related errors* account for a substantial proportion of the total detected instances. These recurring issues contribute significantly to the overall accessibility burden observed in universities and municipalities, while similar patterns are present, though at lower magnitude, in healthcare websites.

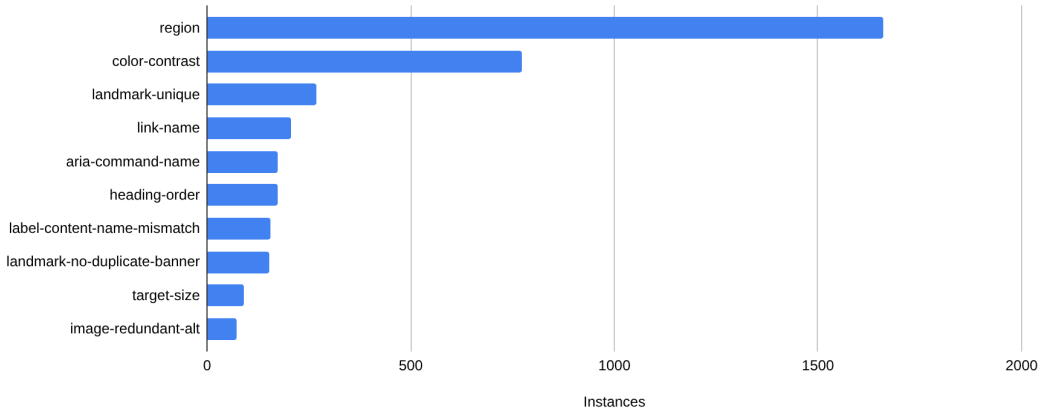


Figure 5.1. Most frequently occurring accessibility issue instances across the evaluated websites

This uneven distribution indicates that accessibility barriers are not uniformly present across domains. Instead, certain domains exhibit higher concentrations of detectable issues, suggesting variation in accessibility burden across institutional contexts.

5.2.3. Diversity

In addition to total issue counts, the number of unique issues provides insight into the diversity of accessibility problems across domains. Universities exhibit the highest number of unique issues (30), indicating a broader range of accessibility challenges.

Healthcare websites show a relatively high number of unique issues (27) despite a lower total issue count. Municipalities, in contrast, exhibit fewer unique issue types (24) while maintaining a high total number of issue instances.

This distinction suggests different underlying patterns of accessibility barriers. High issue counts combined with lower diversity, as observed in municipalities, indicate the presence of repeated or systemic issues affecting multiple pages. In contrast, domains with higher diversity reflect a broader range of distinct accessibility problems.

5.2.4. Severity Distribution

The distribution of accessibility issues across severity levels provides additional insight into the extent of accessibility barriers. As shown in Figure 5.2, accessibility issues are present across all severity levels in each domain, indicating that detected barriers range from minor to more critical concerns.

Moderate-severity issues instances constitute the largest proportion of detected issues across all domains, with particularly high counts observed in universities (1,093) and municipalities (808). This suggests that a significant share of accessibility barriers relates to issues of moderate impact. Examples of recurring moderate-severity issues include region, landmark-unique, and page-has-heading-one related detections, which are associated with navigation structure, page organization, and orientation support.

At the same time, a substantial number of serious issues instances are also observed, particularly in municipalities (934) and universities (607), indicating the presence of barriers with potentially greater impact on user interaction. Healthcare websites exhibit lower counts across severity levels, but still demonstrate the presence of both moderate and serious issues. Recurring serious-severity issues include color-contrast, link-name, heading-order, and label-content-name-mismatch related detections, which may affect readability, navigation clarity, and interaction with interface elements. Healthcare websites exhibit lower counts across severity levels, but still demonstrate the presence of both moderate and serious issues.

Critical issues instances are less frequent overall, but are present in all domains. Universities exhibit the highest number of critical issues (114), followed by healthcare (61) and municipalities (15). This indicates that, although less common, higher-impact accessibility barriers remain present across the evaluated websites. Examples of recurring critical-severity issues include aria-allowed-attr, image-alt, label, and button-name related detections, which are associated with compatibility, non-text content accessibility, and accessible interaction elements.

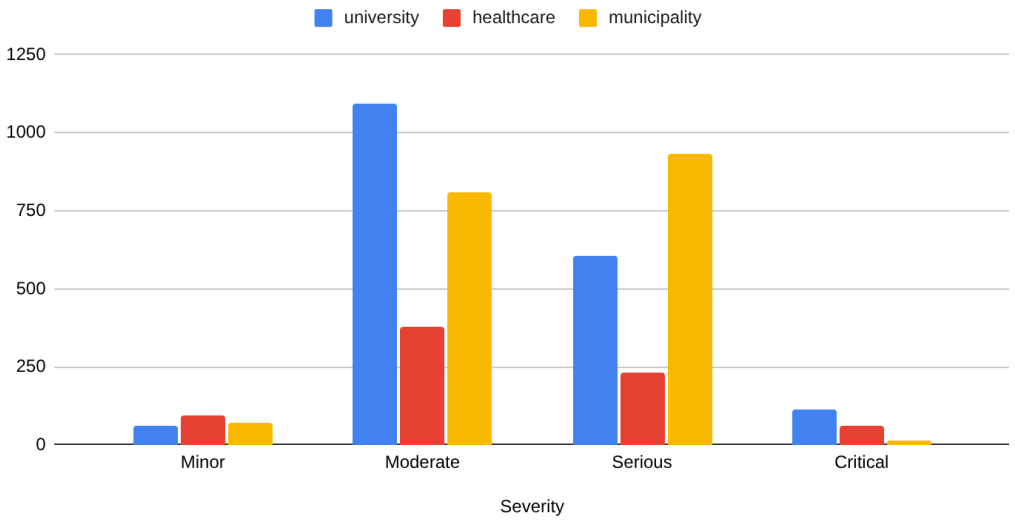


Figure 5.2. Distribution of accessibility issues instances by severity across domains

5.2.5. Normalized Accessibility Burden

To support a more comparable assessment across domains, accessibility issue counts were normalized based on the number of evaluated pages. This allows differences in accessibility burden to be interpreted independently of dataset size.

As shown in Figure 5.3, the number of issue instances per page varies notably across domains. Universities exhibit the highest issue density at approximately 5.21 issues per page, followed closely by municipalities with 5.07 issues per page. In contrast, healthcare websites show a substantially lower issue density of approximately 2.13 issues per page

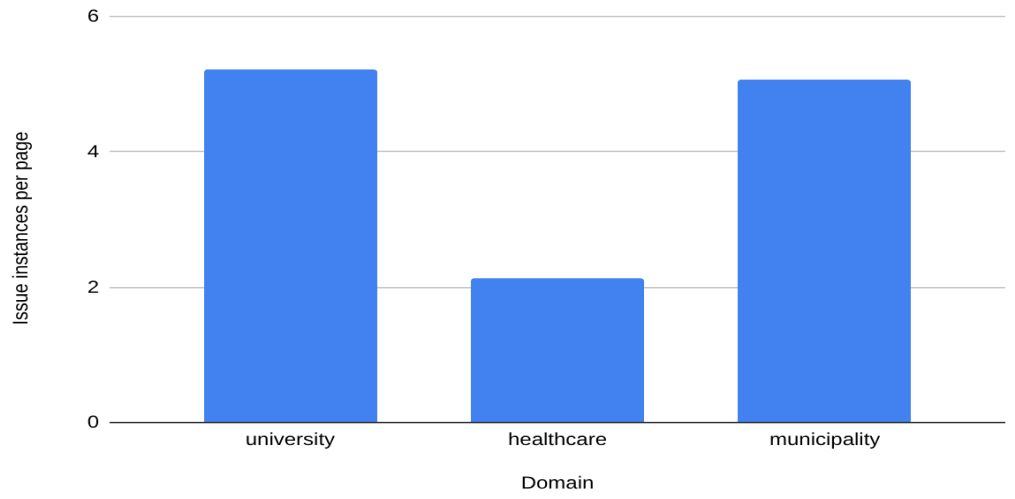


Figure 5.3. Average number of accessibility issue instances per page across domains.

These findings indicate that accessibility barriers are more densely concentrated within university and municipal websites, while healthcare websites exhibit a lower frequency of issues at the page level. This suggests that the differences observed in total issue counts are not solely a result of dataset scale, but also reflect variation in the density of accessibility barriers across domains.

5.2.6. Summary

Taken together, these findings indicate that accessibility barriers are both widespread and unevenly distributed across the evaluated websites. The presence of a large number of issue instances reflects a substantial accessibility burden, while differences in domain distribution and issue diversity highlight variation in how accessibility barriers are distributed across institutional contexts.

5.3. RQ2: Types of Accessibility Barriers and Persona Interpretation

To move beyond the overall extent of accessibility barriers, this section examines the types of issues identified and their potential impact on different user groups. Accessibility findings are analyzed through issue categories, impact groups, and persona-based interpretations, allowing for a structured understanding of how accessibility barriers are distributed and experienced.

5.3.1. Distribution of Accessibility Issue Categories

Accessibility issues are not evenly distributed across categories. Instead, the results show a strong concentration of issue instances within a small number of categories.

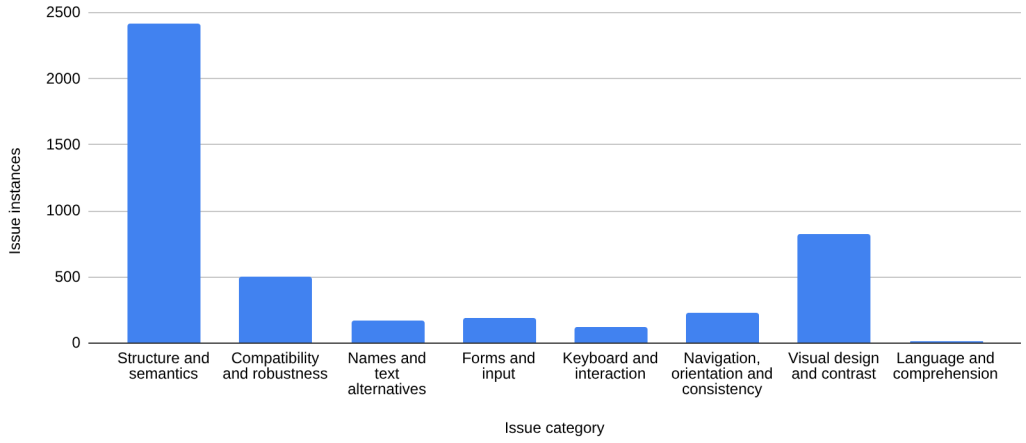


Figure 5.4. Distribution of accessibility issue instances by category.

As shown in Figure 5.4, Structure and semantics constitutes the most dominant category, with 2,414 issue instances, representing a substantial proportion of all detected accessibility issues. This dominance is primarily driven by high-frequency structural

issues such as *region*, *heading order*, and *landmark-related errors*, which affect how content is organized and interpreted.

The second largest category is Visual design and contrast, with 821 issue instances, largely influenced by issues such as *color contrast*. These issues affect the visual perception and readability of content.

Compatibility and robustness also represents a notable category, with 500 issue instances, reflecting issues related to ARIA attributes and compatibility with assistive technologies.

Other categories, including Navigation, orientation and consistency (229 instances), Forms and input (192 instances), and Names and text alternatives (171 instances), contribute to the overall distribution but at lower levels of frequency. Categories such as Keyboard and interaction (124 instances) and Language and comprehension (18 instances) are comparatively less represented.

These findings indicate that accessibility barriers are not uniformly distributed across categories, but are instead heavily concentrated in structural and visual aspects of web design.

5.3.2. Impact Group Distribution

To further understand how accessibility barriers affect users, issue instances were analyzed according to their associated impact groups.

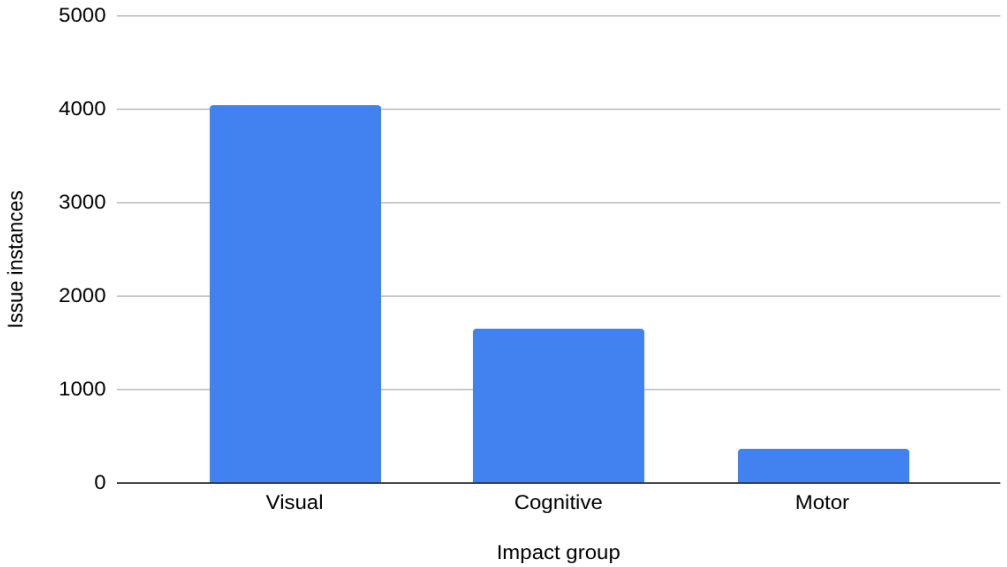


Figure 5.5. Distribution of accessibility issue instances by impact groups.

As shown in Figure 5.5, accessibility issues are strongly concentrated within the visual impact groups, with 4,038 issue instances, representing the vast majority of detected

issues. This reflects the high prevalence of issues related to structure, semantics, and visual presentation, which influence how content is perceived by users and interpreted by assistive technologies.

The cognitive impact groups also represent a substantial portion of accessibility issues, with 1,650 instances. These issues are associated with challenges in content comprehension, clarity, and consistency, often overlapping with structural and visual deficiencies.

In contrast, the motor impact groups are comparatively less represented, with 370 issue instances, primarily related to interaction and input mechanisms such as keyboard accessibility and control usability.

These findings indicate that accessibility barriers are primarily concentrated in areas affecting perception and understanding, while issues related to physical interaction are less prevalent within the evaluated dataset.

This distribution aligns with the dominance of structure- and presentation-related issue categories identified earlier, which primarily affect visual perception and cognitive processing.

5.3.3. Persona-Based Interpretation

To translate technical accessibility findings into user-relevant perspectives, issue instances were analyzed according to their associated personas.

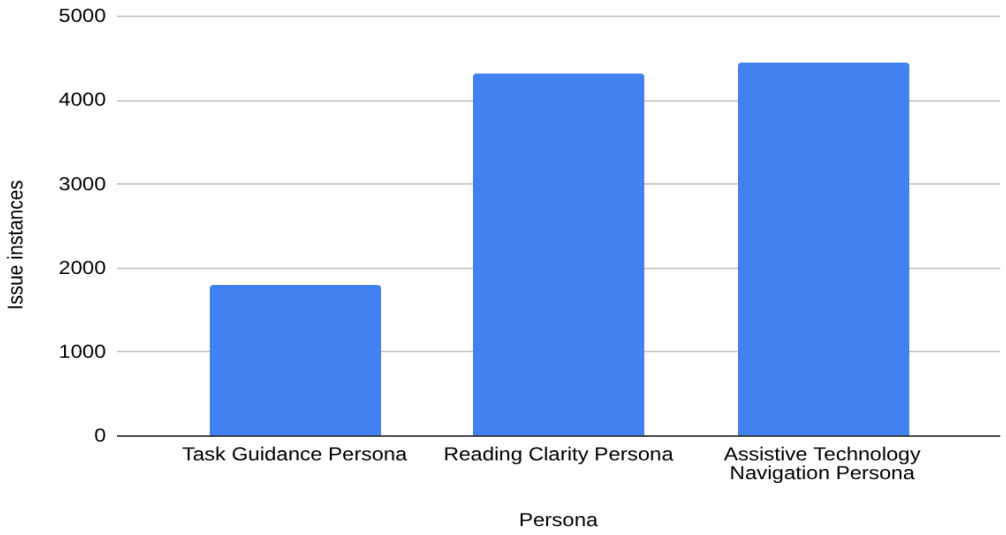


Figure 5.6. Distribution of accessibility issue instances by persona relevance.

As shown in Figure 5.6, accessibility issues are most strongly associated with the Assistive Technology Navigation Persona (4,451 instances) and the Reading Clarity Persona (4,315 instances), both of which account for a substantial proportion of detected issues. This indicates that a large share of accessibility barriers affects users who rely on assistive technologies for navigation, as well as users who depend on clear and well-structured content for comprehension.

The Task Guidance Persona is also affected, with 1,806 issue instances, but to a notably lower extent compared to the other personas. These issues are primarily associated with interaction flows, input handling, and task completion processes.

These findings indicate that accessibility barriers are primarily concentrated in areas affecting navigation and content comprehension, while task-oriented interaction challenges represent a secondary but still relevant dimension.

This distribution aligns with the earlier findings on issue categories and impact groups, where structure- and presentation-related issues were shown to dominate and primarily affect visual perception and cognitive processing.

5.4. RQ3: Comparative Analysis of Accessibility Barriers Across Domains

5.4.1. Category Distribution Across Domains

To examine how accessibility barriers differ across domains, issue instances were analyzed by category within each domain.

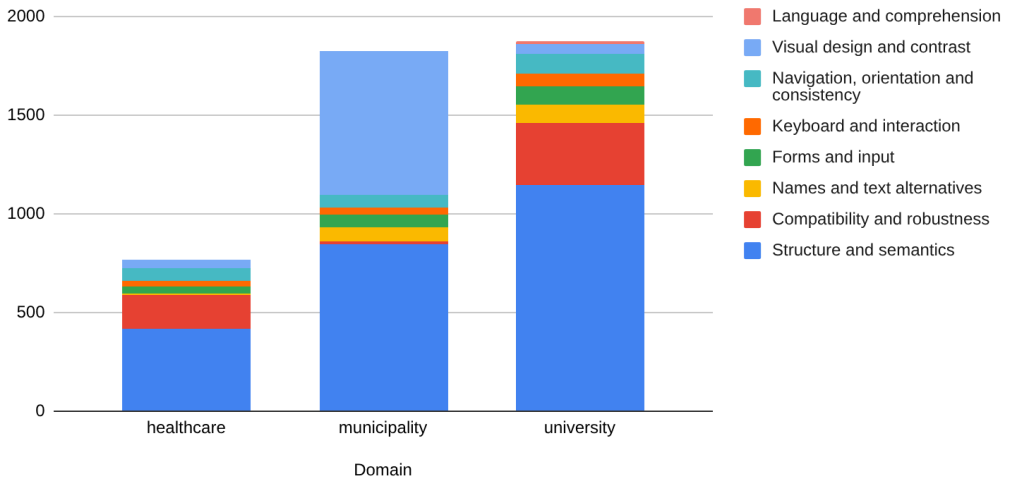


Figure 5.7. Distribution of accessibility issue instances across categories by domain.

As shown in Figure 5.7, clear differences emerge in how accessibility issues are distributed across categories within each domain. While Structure and semantics

represents the largest category across all domains, its relative contribution varies, with universities exhibiting the highest number of instances (1,150), followed by municipalities (843) and healthcare (421).

A notable divergence is observed in the Visual design and contrast category, which is highly prominent in municipalities (730 instances) but comparatively limited in universities (48) and healthcare (43). This indicates that visual accessibility issues are disproportionately concentrated within municipal websites.

Differences are also evident in compatibility and robustness, where universities (314 instances) and healthcare (167 instances) show higher representation compared to municipalities (19 instances), suggesting variation in technical implementation practices across domains.

Other categories, including Navigation, Forms and input, and Names and text alternatives, are more evenly distributed, although still contributing to the overall accessibility burden

These findings indicate that accessibility barriers differ not only in magnitude across domains, but also in their categorical composition, reflecting distinct accessibility profiles within each domain.

5.4.2. Persona Distribution Across Domains

To further examine how accessibility barriers differ across domains from a user perspective, issue instances were analyzed according to persona relevance within each domain.

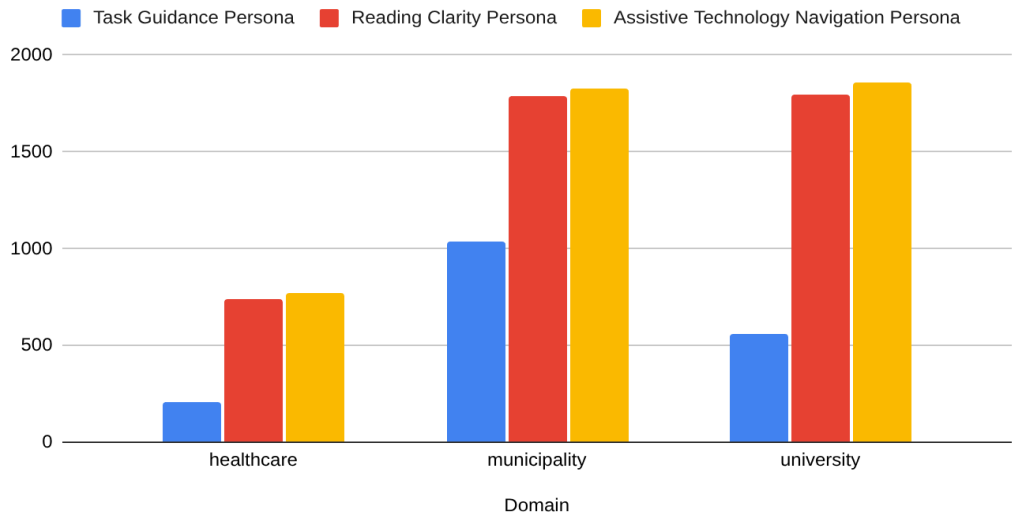


Figure 5.8. Distribution of accessibility issue instances by persona across domains.

As shown in Figure 5.8, the Assistive Technology Navigation Persona and the Reading Clarity Persona represent the most affected user groups across all domains. Universities and municipalities exhibit similarly high numbers of issue instances for both personas, with universities showing 1,858 and 1,793 instances, and municipalities showing 1,826 and 1,786 instances, respectively. Healthcare websites display lower counts for both personas (767 and 736 instances), but maintain a similar distribution pattern.

A notable divergence is observed for the Task Guidance Persona, where municipalities exhibit a substantially higher number of issue instances (1,037) compared to universities (563) and healthcare websites (206). This indicates that task-related accessibility challenges are disproportionately concentrated within municipal websites.

Overall, these findings suggest that while accessibility barriers affecting navigation and content comprehension are consistently prevalent across domains, task-oriented interaction barriers vary more significantly, contributing to distinct accessibility profiles within each domain.

This pattern is consistent with the earlier category-level findings, where interaction- and form-related issues were more prominent in municipal websites.

5.4.3. Summary of Domain Differences

Taken together, the findings reveal distinct patterns in how accessibility barriers are distributed across domains, indicating the presence of different accessibility profiles.

Universities and municipalities exhibit the highest overall accessibility burden, but differ in their internal composition. Universities are characterized by a strong concentration of structure- and compatibility-related issues, which contribute to high levels of impact on both the Reading Clarity Persona and the Assistive Technology Navigation Persona. In addition, universities exhibit the highest number of critical issues, indicating the presence of barriers with potentially greater impact.

Municipal websites, while similar in overall issue volume, display a markedly different distribution. In particular, they exhibit a substantial concentration of visual design and contrast issues, as well as a significantly higher number of issues affecting the Task Guidance Persona. This domain also shows the highest number of serious issues, suggesting that accessibility barriers are more frequently concentrated at higher impact levels.

Healthcare websites, in contrast, show a consistently lower number of issue instances across all categories and personas. Their distribution is comparatively more balanced, without extreme concentrations in specific categories. While accessibility barriers are still present, their overall profile suggests both a lower intensity and a lower concentration of high-severity issues.

Across all domains, a common pattern emerges in which issues affecting navigation and content comprehension—as reflected in the Reading Clarity and Assistive Technology Navigation personas—are consistently prevalent. However, differences in category composition and severity levels contribute to distinct accessibility characteristics for each domain.

6. Discussion

6.1. Extent of Accessibility Barriers

The findings show that accessibility barriers are present across all evaluated public-sector domains, indicating that detectable accessibility issues remain a persistent feature of Swedish public institution websites. This aligns with prior research showing that public-sector websites continue to exhibit accessibility barriers despite established standards, legislation, and policy frameworks (Alajarmeh, 2021; Vollenwyder et al., 2023; Jonsson, Gustavsson, & Gulliksen, 2023).

The presence of 4,469 issue instances across the evaluated websites suggests that accessibility barriers are not isolated occurrences, but a recurring condition across the sampled public-service contexts. This supports the view that formal accessibility requirements, such as WCAG and public-sector accessibility legislation, do not automatically translate into fully accessible digital services. Rather, the findings reinforce the implementation gap identified in prior research, where compliance frameworks provide necessary guidance but do not by themselves ensure accessible outcomes in practice (Inal et al., 2022; Jonsson, Gustavsson, & Gulliksen, 2023).

The normalized issue-density results further strengthen this interpretation. Universities and municipalities show substantially higher issue densities than healthcare websites, with universities and municipalities both exceeding five issue instances per page, while healthcare websites show a lower density. This indicates that the higher issue counts in universities and municipalities are not simply a result of larger evaluated samples, since the page scope was balanced across domains. However, differences in website structure, content complexity, and interaction design may also contribute to variation in issue density across domains. For example, university and municipal websites often contain broader informational structures and more complex navigation and service-related content, which may increase the likelihood of detectable accessibility barriers.

Severity distribution adds another important dimension to the interpretation of accessibility extent. The findings show that issues are not limited to minor concerns, but include substantial numbers of moderate, serious, and critical issues. This matters because accessibility burden is not only a question of how many issues exist, but also how much potential impact they may have on users. In particular, the presence of serious and critical issues indicates that some barriers may meaningfully affect users' ability to perceive, navigate, or interact with digital public services.

Taken together, the results addressing RQ1 indicate that accessibility remains a significant issue across the evaluated public-sector websites. They also demonstrate the value of examining accessibility burden through multiple analytical dimensions, including total issue instances, issue density, unique issue definitions, and severity. This multi-dimensional perspective provides a more nuanced understanding than relying solely

on compliance scores or raw issue counts, and supports the study's barrier-oriented approach to accessibility evaluation.

6.2. Accessibility Barrier Types and User Impact

The findings related to barrier types reveal a clear concentration of accessibility issues within a limited set of categories, most notably Structure and semantics, Visual design and contrast, and Compatibility and robustness. This pattern aligns with prior research identifying semantic structure, navigation, and visual presentation as recurring problem areas in public-sector websites (Campoverde-Molina et al., 2023; Inal et al., 2022; Kesswani & Kumar, 2022). The dominance of these categories suggests that accessibility challenges are not randomly distributed, but instead emerge from foundational aspects of web design and implementation.

The strong representation of structure and semantics issues is particularly significant. These issues affect how content is organized, interpreted, and navigated, both visually and through assistive technologies. This supports existing literature emphasizing the importance of semantic markup and logical document structure as core requirements for accessibility (W3C, 2024). The findings therefore reinforce the view that structural correctness is not a secondary technical detail, but a fundamental prerequisite for accessible interaction.

The distribution across impact groups further clarifies how these barriers affect users. The dominance of the visual and cognitive impact groups indicates that most accessibility issues influence perception and comprehension rather than purely physical interaction. This is consistent with accessibility research highlighting that many barriers arise from how information is presented and structured, rather than from input mechanisms alone (Henry, Abou-Zahra, & Brewer, 2014; Lazar, Goldstein, & Taylor, 2015). The comparatively lower representation of the motor impact groups suggests that interaction-related issues are present but less central within the observed dataset.

The persona-based interpretation adds an additional layer of understanding by translating these technical patterns into user-relevant perspectives. The findings show that the Reading Clarity Persona and the Assistive Technology Navigation Persona are most strongly affected across the dataset, while the Task Guidance Persona is affected to a lesser extent. This aligns with the observed dominance of structural and visual issues, as both personas are directly linked to content comprehension and navigation through assistive technologies.

Importantly, this result supports the usefulness of the persona framework as an interpretive tool. Rather than treating accessibility issues as abstract technical violations, the persona mapping provides a structured way to infer which types of user interaction are most likely to be affected. At the same time, this interpretation remains consistent with the methodological boundary established in the framework: persona-based insights

represent analytical approximations rather than direct measurements of user experience (Turner & Turner, 2011; Marshall et al., 2015).

Taken together, the findings for RQ2 suggest that accessibility barriers in the evaluated websites are primarily concentrated in areas affecting content structure, visual clarity, and assistive-technology navigation. This reinforces the argument that accessibility challenges are deeply embedded in how digital content is designed and structured, rather than being limited to isolated usability issues.

6.3. Cross-Domain Differences in Accessibility Barriers

The comparative analysis across domains reveals that accessibility barriers are not only widespread, but also structured differently depending on institutional context. While universities, municipalities, and healthcare websites all exhibit accessibility issues, the findings show that these domains differ in both the composition and severity of their accessibility barriers. This supports the argument that accessibility cannot be understood solely as a uniform compliance problem, but must be examined as a domain-sensitive phenomenon.

Universities and municipalities exhibit similarly high overall accessibility burdens, yet their internal barrier profiles differ in important ways. Universities are characterized by a strong concentration of structure- and compatibility-related issues, which are closely associated with barriers affecting navigation and interaction through assistive technologies. This aligns with prior research highlighting persistent issues in semantic markup and assistive technology compatibility (Campoverde-Molina et al., 2023; Inal et al., 2022). The presence of a higher number of critical issues in universities further suggests that, in addition to frequency, some barriers may have a higher potential impact on user interaction.

Municipal websites, in contrast, show a different pattern. While they also exhibit a high overall number of issues, they are distinguished by a substantial concentration of visual design and contrast issues, as well as a significantly higher number of issues affecting the Task Guidance Persona. This indicates that accessibility challenges in municipal contexts are more strongly related to visual clarity and task-oriented interaction. Although prior studies have identified contrast and usability issues as common accessibility barriers (Kesswani & Kumar, 2022; Akgül, 2021), the present findings extend this understanding by showing how these issues can be disproportionately concentrated within a specific domain.

Healthcare websites display a comparatively lower accessibility burden across all analytical dimensions. They show fewer issue instances, lower issue density, and a more balanced distribution across categories and personas. This pattern suggests that accessibility barriers in healthcare websites are less concentrated and potentially less structurally dominant. While prior literature often reports accessibility challenges in healthcare systems (Valtolina & Fratus, 2022), the findings here indicate that, within the

studied sample, healthcare websites exhibit a different accessibility profile compared to universities and municipalities. This difference may reflect variations in content structure, service focus, or development practices, although such factors are not directly measured in this study.

Across all domains, a consistent pattern emerges in which barriers affecting navigation and content comprehension remain prevalent. This reflects the broader importance of structural clarity and semantic correctness in web accessibility, as emphasized in both WCAG guidelines and prior research (W3C, 2024; Henry, Abou-Zahra, & Brewer, 2014). However, the way these barriers are combined with other issue types varies across domains, resulting in distinct accessibility profiles.

Taken together, these findings contribute to answering RQ3 by demonstrating that accessibility barriers are not only prevalent across public-sector websites, but also differ systematically in structure and impact across institutional domains. While prior research has often focused on single-domain evaluations or descriptive issue inventories, the use of a consistent analytical framework across universities, municipalities, and healthcare websites makes it possible to identify domain-specific accessibility profiles and variations in accessibility burden.

6.4. Methodological Reflections

The methodological design of this study reflects a deliberate balance between scalability, consistency, and interpretability in the analysis of web accessibility barriers. The use of automated evaluation tools enabled systematic data collection across a relatively large sample of websites, supporting cross-domain comparison under controlled conditions. This aligns with prior research emphasizing the necessity of automated methods for large-scale accessibility analysis (Campoverde-Molina et al., 2023; Inal et al., 2022).

At the same time, the limitations of automated evaluation remain significant. As highlighted in the literature, automated tools can only detect a subset of WCAG criteria and are unable to fully capture context-dependent or interaction-based accessibility barriers (Fischer et al., 2025; Vigo, Brown, & Conway, 2013). The findings of this study should therefore be interpreted as structured indicators of accessibility barriers, rather than complete representations of user experience. This limitation reinforces the importance of cautious interpretation and supports the decision to treat issue instances as analytical signals rather than definitive usability outcomes.

To address some of these limitations, the study introduced a taxonomy-driven analytical framework that organizes raw tool outputs into issue definitions, categories, and impact groups. This approach responds to a common limitation in prior research, where accessibility findings are often reported as isolated technical issues without structured interpretation (Valtolina & Fratus, 2022). By normalizing heterogeneous tool outputs into consistent analytical units, the taxonomy enabled meaningful comparison across domains and supported abstraction from low-level detections to higher-level barrier patterns.

The inclusion of a persona-based interpretive layer further extends the analytical approach. Rather than treating accessibility findings as user-neutral, the persona framework provides a structured mechanism for linking technical issues to likely user-impact perspectives. This aligns with literature suggesting that personas can enhance the interpretability of accessibility findings when direct user testing is not feasible (Marshall et al., 2015). However, consistent with prior warnings in persona research, this study treats persona mappings as analytical approximations, not as direct representations of real user experiences (Turner & Turner, 2011). This boundary is important to avoid overgeneralization and to maintain methodological transparency.

Another important methodological contribution is the use of multi-level measurement, distinguishing between issue-instance counts and unique issue-definition counts. This separation allowed the analysis to capture both the intensity and diversity of accessibility barriers, reducing the risk of misinterpretation that can arise when different analytical units are conflated. Similarly, the inclusion of normalized metrics, such as issue instances per page, supported fair comparison across domains by controlling for differences in dataset scale.

Overall, the methodological approach demonstrates how automated accessibility evaluation can be extended beyond compliance-oriented reporting toward structured, comparative, and interpretable analysis. While the approach does not overcome all limitations of automated methods, it provides a framework for making such data more analytically meaningful and relevant for both research and practice.

6.5. Practical Implications and Recommendations

The findings of this study have several implications for the design, development, and governance of accessible public-sector websites. Some implications reinforce established accessibility principles and WCAG-aligned best practices, while others emerge more directly from the taxonomy-driven and persona-informed analytical approach developed in this study.

6.5.1. General Accessibility Recommendations

The strong concentration of accessibility issues within structure and semantics highlights the foundational role of semantic correctness in web accessibility. Issues related to headings, landmarks, and content structure affect both navigation and content comprehension, particularly for users relying on assistive technologies. Public-sector organizations should therefore prioritize the correct use of semantic HTML, heading hierarchies, landmarks, and meaningful labels across digital services.

The prominence of visual design and contrast issues, particularly within municipal websites, further indicates that accessibility is strongly influenced by design-level decisions. These findings suggest that accessibility should not be treated solely as a technical compliance concern, but also as an integral aspect of visual and interaction

design. Organizations should therefore integrate accessibility requirements into design systems and visual guidelines to support readability and visual clarity across pages and services.

The observed concentration of task-related barriers, especially in municipal contexts, highlights the importance of accessible interaction design and service workflows. Accessibility challenges in this area are closely linked to how users navigate forms, understand instructions, and complete tasks. Organizations should therefore improve form usability through clearer labeling, understandable instructions, meaningful error feedback, and more predictable interaction flows.

The presence of compatibility and robustness issues, particularly in university and healthcare websites, underscores the importance of technically sound implementations supporting reliable interaction with assistive technologies. These findings indicate that accessibility is influenced not only by visible interface elements, but also by underlying technical practices affecting how content is interpreted across different user agents.

More broadly, the findings suggest that accessibility should be treated as a continuous quality attribute rather than a one-time compliance activity. Organizations should integrate accessibility evaluation into development workflows and conduct regular reviews following updates or system changes to support sustained accessibility improvement over time.

6.5.2. Study-Specific Recommendations

Beyond reinforcing established accessibility practices, the findings of this study suggest the value of more structured and data-driven approaches to accessibility evaluation and prioritization. The results demonstrate that accessibility barriers are not evenly distributed, but instead cluster around recurring structural, semantic, and interaction-related issue patterns across domains.

This suggests that organizations may benefit from moving beyond isolated issue correction toward more systematic approaches to accessibility management. Rather than treating accessibility findings as independent technical defects, issues can be categorized, analyzed for recurring patterns, and prioritized based on normalized severity and potential user impact. The taxonomy- and persona-informed analytical framework developed in this study demonstrates how accessibility evaluation data can support more structured and interpretable accessibility analysis.

The findings also suggest that accessibility governance approaches may need to be adapted to institutional context. Universities and municipalities exhibited higher concentrations of structural and navigation-related barriers, potentially reflecting differences in website complexity, decentralized content management practices, and variation in digital service structures. This indicates that accessibility improvement strategies may benefit from accounting for domain-specific conditions rather than relying solely on uniform compliance procedures.

For policymakers and public-sector administrators, the findings highlight limitations in relying exclusively on compliance-oriented accessibility reporting. While WCAG-based regulations provide necessary normative guidance, the results suggest that additional emphasis should be placed on implementation support, structured monitoring practices, and analytical approaches capable of identifying recurring accessibility burden patterns across domains.

For developers and technical teams, the findings demonstrate the practical value of prioritizing accessibility work based on recurring issue frequency, normalized severity, and likely user impact. By grouping accessibility findings into structured analytical categories and examining their relation to different interaction perspectives, organizations may be better positioned to allocate resources efficiently and target accessibility improvements with greater practical impact.

7. Conclusion

7.1. Conclusions

This thesis set out to analyze accessibility barriers across Swedish public-sector websites and to examine how these barriers are distributed, structured, and differentiated across institutional domains. The findings indicate that accessibility remains a significant and persistent challenge, despite established standards and regulatory frameworks.

In relation to RQ1, the study demonstrates that accessibility barriers are widespread across all evaluated domains. A substantial number of issue instances were identified, and normalized metrics confirm that these barriers are not simply a function of dataset size, but reflect meaningful differences in issue density across domains. Furthermore, the presence of moderate, serious, and critical issues indicates that accessibility challenges extend beyond minor technical concerns and include barriers with potentially significant impact on user interaction.

With regard to RQ2, the findings show that accessibility barriers are not evenly distributed across types, but are highly concentrated within a limited number of categories. In particular, issues related to structure and semantics, visual design and contrast, and compatibility and robustness dominate the dataset. These barriers primarily affect visual perception and cognitive processing, as reflected in the strong representation of the Reading Clarity Persona and the Assistive Technology Navigation Persona. This indicates that accessibility challenges are deeply embedded in how content is structured, presented, and interpreted.

In addressing RQ3, the study demonstrates that accessibility barrier profiles differ across domains. While universities and municipalities exhibit similarly high overall accessibility burdens, they differ in their internal composition. Universities are characterized by structural and compatibility-related issues, along with a higher presence of critical barriers, whereas municipalities show a strong concentration of visual and task-oriented interaction issues. Healthcare websites, in contrast, display a lower and more balanced distribution of accessibility barriers. These findings indicate that accessibility challenges are not uniform across public-sector contexts, but instead reflect domain-specific patterns.

Taken together, the results show that accessibility in public-sector websites cannot be fully understood through isolated compliance measures or issue counts alone. Instead, a multi-dimensional perspective—considering extent, type, and distribution of barriers across domains and user perspectives—is necessary to capture the complexity of accessibility challenges.

Based on these findings, the study highlights several key priorities for improving accessibility in public-sector websites. Efforts should focus on strengthening structural and semantic correctness, improving visual clarity and contrast, and enhancing task-oriented interaction design, particularly in form-based services. In addition, ensuring

compatibility with assistive technologies remains essential for supporting diverse user needs. Taken together, these priorities suggest that accessibility should be addressed as an integrated and continuous aspect of design and development practice, rather than as a one-time compliance activity.

7.2. Contributions

This thesis contributes a structured, taxonomy-driven approach for analyzing web accessibility barriers across public-sector domains. By transforming heterogeneous outputs from automated evaluation tools into normalized issue definitions, categories, and user-impact dimensions, the study enables consistent and interpretable cross-domain comparison. In addition, a persona-informed interpretive layer is introduced to link technical accessibility issues to likely patterns of user impact, supporting more meaningful analysis beyond compliance-based reporting. Applied to Swedish public-sector websites, the approach demonstrates that accessibility barriers are not only widespread, but also systematically structured and domain-dependent. Building on this, the thesis makes several specific contributions to the study of web accessibility in public-sector digital services.

First, the study provides an empirical contribution by offering a structured analysis of accessibility barriers across Swedish public institution websites. While prior research has frequently examined accessibility within single domains or individual organizations, this study applies a consistent evaluation framework across multiple domains, enabling systematic cross-domain comparison. This contributes to a more comprehensive understanding of how accessibility barriers are distributed and how they differ across institutional contexts.

Second, the thesis contributes methodologically by introducing a taxonomy-driven analytical approach to accessibility evaluation. By normalizing automated tool outputs into issue definitions, categories, and impact groups, the study demonstrates how raw accessibility findings can be transformed into structured, comparable, and interpretable analytical units. This addresses a common limitation in prior research, where accessibility results are often reported as isolated technical issues without higher-level organization.

Third, the study extends existing approaches by incorporating a persona-informed interpretive layer. This enables accessibility findings to be connected to user-relevant perspectives, linking technical barrier patterns to likely user-impact implications. While personas are used as analytical abstractions rather than direct representations of user experience, their inclusion enhances the interpretability of accessibility data and supports more meaningful comparison across domains.

Fourth, the thesis contributes by applying a multi-dimensional analytical model that combines measures of accessibility extent, barrier type, and cross-domain variation. By distinguishing between issue-instance counts and unique issue definitions, and by incorporating normalized metrics and severity levels, the study provides a more nuanced

understanding of accessibility barriers than approaches based solely on compliance scores or raw issue counts.

Finally, the study offers a practical contribution by deriving actionable recommendations grounded in empirical findings. These recommendations demonstrate how structured accessibility analysis can inform prioritization and support more effective accessibility improvement strategies in public-sector contexts.

7.3. Limitations

This study has several limitations that should be considered when interpreting the findings.

First, the analysis is based on automated accessibility evaluation tools, which are known to detect only a subset of WCAG-defined accessibility issues. As highlighted in prior research, automated tools cannot fully capture context-dependent, interaction-based, or content-quality-related barriers (Fischer et al., 2025; Vigo, Brown, & Conway, 2013). The study also does not include manual validation of automated findings, which may affect the precision of certain issue classifications and severity interpretations. As a result, the findings should be interpreted as indicators of accessibility barriers, rather than a complete assessment of user experience.

Second, the study is limited to a selected sample of Swedish public institution websites across three domains. Although the sampling strategy ensured balanced representation, the findings are not intended to be generalized to all public-sector websites. Instead, they reflect the accessibility characteristics of the specific sample and evaluation scope defined in this study.

Third, the evaluation is based on a fixed page selection strategy, with a limited number of pages per website. While this approach supports comparability across domains, it may not capture the full variability of accessibility barriers within each website. Some pages or services not included in the sample may exhibit different accessibility characteristics.

Fourth, the persona-based interpretation represents an analytical abstraction rather than direct observation of real user behavior. Personas are used as structured interpretive lenses to infer likely user-impact patterns, but they do not replace user testing or empirical evidence of lived user experience. Consequently, persona-related findings should be understood as analytically derived approximations, not definitive statements about how users experience accessibility barriers.

Finally, while the study applies a taxonomy-driven normalization approach to improve comparability and interpretability, this process involves researcher-defined mappings between issue definitions, categories, and impact groups. Although these mappings are grounded in established frameworks and literature, they introduce a degree of analytical subjectivity.

7.4. Future Research

The findings and limitations of this study suggest several directions for future research.

First, future studies could incorporate manual accessibility evaluation and user-based testing to complement automated analysis. While automated tools enable scalable detection of accessibility barriers, they cannot fully capture context-dependent or interaction-based issues. Combining automated methods with user testing, particularly involving users with disabilities, would provide deeper insight into how detected barriers affect real-world interaction and user experience.

Second, further research could expand the scope and scale of analysis by including a larger and more diverse set of public-sector websites, both within Sweden and across different national contexts. This would enable broader generalization of findings and support comparative analysis of accessibility practices across regulatory and institutional environments.

Third, longitudinal studies could examine how accessibility evolves over time, particularly in response to regulatory requirements, organizational initiatives, or technological changes. Such studies could provide insight into whether accessibility improvements are sustained and how barrier patterns shift as digital services develop.

Fourth, future research could explore domain-specific factors influencing accessibility barriers, such as organizational practices, content complexity, governance structures, and development workflows. While this study identifies differences between domains, it does not directly measure the underlying causes of these differences. Investigating these factors would support a deeper understanding of why accessibility profiles vary across institutional contexts.

Fifth, the taxonomy-driven and persona-informed analytical approach introduced in this study could be further refined and validated. Future work could explore alternative taxonomy structures, expand impact-group definitions, or empirically evaluate persona mappings through user studies. This would strengthen the interpretive link between technical accessibility findings and user experience.

Finally, research could investigate how structured accessibility analysis can be integrated into practical development and governance processes, including design systems, continuous monitoring pipelines, and organizational accessibility strategies. This would help bridge the gap between accessibility evaluation and actionable implementation in real-world settings.

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Appendices

Appendix A: Report on the use of generative AI in master thesis

Purpose of generative AI use

The main AI tool used in this thesis was ChatGPT. It played a small, helpful role during the research and writing. Early on, ChatGPT helped with brainstorming by exploring possible research ideas and refining initial thoughts. It was also used to make the language clearer by suggesting different ways to say things and making the text easier to read. Later, ChatGPT helped with proofreading by finding grammar mistakes and improving the overall quality of the writing.

Extent of generative AI use

ChatGPT was used in this thesis only for certain helpful tasks. It was not used for the whole research process. Instead, it was used only to generate ideas early on and to improve the language while writing. Importantly, ChatGPT was not used for key academic tasks such as analyzing data, interpreting results, building arguments, or drawing conclusions. The authors did these parts themselves to keep the work original and thorough.

Impact of generative AI use

Using ChatGPT made the writing process faster and helped clarify and improve the thesis's readability. It assisted with first ideas and made expressing thoughts smoother. However, suggestions from the AI needed careful review because they could sometimes be unclear or insufficiently detailed for academic work. So, all ChatGPT results were carefully reviewed and changed when needed. Overall, the tool was a helpful assistant, not a substitute for doing academic work on your own.

Personal contribution

The authors take full responsibility for all main parts of the thesis. The original research idea, along with the design, data collection, analysis, interpretation, and conclusions, was developed by the authors. All arguments and thoughtful reflections in the thesis come from the authors' own judgment and analysis. Any suggestions from ChatGPT were carefully reviewed and used only when suitable, ensuring the authors' own voice and academic work remained at the center of the thesis.

Ethical considerations of generative AI use

ChatGPT was used in accordance with university rules and academic honesty. No plagiarism was done, and all parts of the thesis are the authors' own work. Also, no private or sensitive information was given to the AI. All AI results were carefully checked to ensure they were correct, trustworthy, and in compliance with academic rules.

Reflection on non-use of generative AI

ChatGPT was purposely not used in important parts of the thesis, like analyzing data, understanding results, and drawing conclusions. This choice was made to allow a closer connection with the research and to have full academic responsibility for the analysis. By using our own knowledge and critical thinking in these parts, we improved both the quality and honesty of the thesis.

Appendix B: Description of the work division between the authors

General Description of Collaboration

We worked closely together throughout the thesis. We came up with the research idea together, and we both helped shape the study's focus and goals equally. We planned and organized the work by staying in constant contact and having regular Zoom meetings, which allowed both of us to take part in decision-making. We divided the tasks mainly by splitting the writing, but both shared responsibility for the overall quality and flow of the thesis. During the process, we checked each other's work and gave feedback to ensure the result was clear and well-connected.

Division of Work Across the Thesis Work

The thesis was done through a mix of working together and splitting tasks. We both shared ideas and planned each chapter equally. However, the writing was split up to make the process efficient and clear. One author mainly wrote the introduction, while the literature review was created together by splitting its parts between us. The methodology chapter was also written together, with each of us handling different sections. We both collected the data. The second author mainly wrote the discussion chapter. We worked together equally on the conclusion. In the end, we both edited and revised the entire thesis to ensure it was clear, consistent, and met academic standards.

Individual Contributions

Fardin Hakimi:

I was primarily responsible for writing the theoretical framework, findings and discussion chapters and contributed to the literature review and methodology by writing assigned sections. I also participated actively in data collection and contributed to the development of the conclusion. Furthermore, I reviewed and edited all parts of the thesis to ensure coherence and quality.

Rahmat Surakat:

I mainly wrote the introduction and literature review chapters and contributed to methodology by writing the sections I was assigned. I also took part in collecting data and helped develop the conclusion. Additionally, I reviewed and edited the entire thesis to make sure it was clear and well-written.

Statement of Agreement

Both authors agree that this description correctly shows how the work was divided and each person's contributions to the thesis. All authors have checked and approved the content of this appendix.

Appendix C: Mapping of Issue Definitions, Categories, and Impact Groups

Table C1. Mapping of Issue Definitions to Issue Categories and Impact Groups

Issue	WCAG	Category	Impact groups
accesskeys	-	Keyboard and interaction	Motor
aria-allowed-attr	4.1.2	Compatibility and robustness	Visual
aria-allowed-role	-	Compatibility and robustness	Visual
aria-command-name	4.1.2	Compatibility and robustness	Visual
aria-hidden-focus	4.1.2	Compatibility and robustness	Visual, Motor
aria-input-field-name	4.1.2	Compatibility and robustness	Visual
aria-valid-attr	4.1.2	Compatibility and robustness	Visual
aria-valid-attr-value	4.1.2	Compatibility and robustness	Visual
button-name	4.1.2	Names and text alternatives	Motor, Cognitive
color-contrast	1.4.3	Visual design and contrast	Visual, Cognitive
document-title	2.4.2	Navigation, orientation and consistency	Cognitive
empty-heading	-	Structure and semantics	Visual, Cognitive
frame-title	4.1.2	Names and text alternatives	Cognitive, Visual
heading-order	-	Structure and semantics	Visual, Cognitive
html-has-lang	3.1.1	Language and	Cognitive

		comprehension	
image-alt	1.1.1	Names and text alternatives	Cognitive, Visual
image-redundant-alt	-	Names and text alternatives	Visual, Cognitive
label	4.1.2	Forms and input	Motor, Cognitive
label-content-name-mismatch	2.5.3	Forms and input	Cognitive, Motor
label-title-only	-	Forms and input	Cognitive, Motor
landmark-banner-is-top-level	-	Structure and semantics	Visual
landmark-complementary-is-top-level	-	Structure and semantics	Visual
landmark-contentinfo-is-top-level	-	Structure and semantics	Visual
landmark-main-is-top-level	-	Structure and semantics	Visual
landmark-no-duplicate-banner	-	Structure and semantics	Visual
landmark-no-duplicate-contentinfo	-	Structure and semantics	Visual
landmark-no-duplicate-main	-	Structure and semantics	Visual
landmark-one-main	-	Structure and semantics	Visual
landmark-unique	-	Structure and semantics	Visual
link-in-text-block	1.4.1	Visual design and contrast	Visual, Cognitive
link-name	2.4.4	Navigation, orientation and consistency	Cognitive, Visual
list	1.3.1	Structure and semantics	Cognitive, Visual
listitem	1.3.1	Structure and semantics	Cognitive, Visual
meta-viewport-large	-	Visual design and contrast	Visual

nested-interactive	4.1.2	Compatibility and robustness	Motor
page-has-heading-one	-	Structure and semantics	Cognitive
presentation-role-conflict	-	Compatibility and robustness	Visual
region	-	Structure and semantics	Visual
role-img-alt	1.1.1	Names and text alternatives	Cognitive, Visual
scrollable-region-focusable	2.1.1	Keyboard and interaction	Motor
select-name	4.1.2	Forms and input	Motor, Cognitive
skip-link	-	Navigation, orientation and consistency	Motor, Cognitive
table-fake-caption	1.3.1	Structure and semantics	Cognitive, Visual
target-size	2.5.8	Keyboard and interaction	Motor
td-has-header	1.3.1	Structure and semantics	Cognitive, Visual

Appendix D: Evaluated Websites by Domain and Page Coverage

Table D1. Evaluated Websites by Domain and Page Coverage

Domain	Institution	Website URL	Pages Evaluated	Evaluation Runs
Healthcare	1177	https://www.1177.se/	30	1
Healthcare	Halland County	https://www.regionhalland.se/halsa-och-vard	30	1
Healthcare	Jämtland Härjedalen County	https://www.regionjh.se/halsa--vard	30	1
Healthcare	Jönköping County	https://www.rjl.se/Folkhalsa-och-vard/	30	1
Healthcare	Sörmland County	https://regionsormland.se/halsa-vard/	30	1
Healthcare	Stockholm County	https://www.regionstockholm.se/halsa-sjukvard/	30	1

Healthcare	Swedish Dental and Pharmaceutical Benefits Agency	https://www.tlv.se/	30	1
Healthcare	Swedish eHealth Agency	https://www.ehalsomyndigheten.se/	30	1
Healthcare	Swedish Medical Products Agency	https://www.lakemedelsverket.se/sv	30	1
Healthcare	Swedish National Board of Health and Welfare	https://www.socialstyrelsen.se/	30	1
Healthcare	Swedish Public Health Agency of Sweden	https://www.folkhalsomyndigheten.se/	30	1
Healthcare	Västra Götaland County	https://www.vgregion.se/halsa-och-vard/	30	1
Municipality	Falkenberg Municipality	https://kommun.falkenberg.se/	30	1
Municipality	Gislaved Municipality	https://www.gislaved.se/	30	1
Municipality	Jonkoping Municipality	https://www.jonkoping.se/	30	1
Municipality	Kalmar Municipality	https://kalmar.se/	30	1
Municipality	Karlskoga Municipality	https://karlskoga.se/	30	1
Municipality	Kungsbacka Municipality	https://kungsbacka.se/	30	1
Municipality	Linköping Municipality	https://www.linkoping.se/	30	1
Municipality	Örnsköldsvik Municipality	https://www.ornskoldsvik.se/	30	1

Municipality	Osby Municipality	https://www.osby.se/	30	1
Municipality	Skövde Municipality	https://www.skovde.se/	30	1
Municipality	Umeå Municipality	https://www.umea.se/	30	1
Municipality	Växjö Municipality	https://www.vaxjo.se/	30	1
University	Boras university	https://www.hb.se/	30	1
University	Gothenburg University	https://www.gu.se/	30	1
University	Jonkoping University	https://ju.se/	30	1
University	Linnaeus university	https://lnu.se/	30	1
University	Luleå University of Technology	https://www.ltu.se/	30	1
University	Lund University	https://www.lu.se/	30	1
University	Malmö University	https://mau.se/	30	1
University	Mid Sweden University	https://www.miun.se/	30	1
University	Swedish Defence University	https://www.fhs.se/	30	1
University	Umeå University	https://www.umu.se/	30	1
University	University West	https://www.hv.se/	30	1
University	Uppsala University	https://www.uu.se/	30	1

Appendix E: Accessibility Issue Matrix

Table E1. Accessibility issue matrix showing issue definitions, total instances, normalized severity, affected domains, and associated persona relevance across the dataset.

Issue	Normalized severity	Instances	Affected domains	Affected personas
region	moderate	1660	healthcare, municipality, university	Reading Clarity Persona, Assistive Technology Navigation Persona
color-contrast	serious	773	healthcare, municipality, university	Task Guidance Persona, Reading Clarity Persona, Assistive Technology Navigation Persona
landmark-unique	moderate	269	healthcare, municipality, university	Reading Clarity Persona, Assistive Technology Navigation Persona
link-name	serious	207	healthcare, municipality, university	Task Guidance Persona, Reading Clarity Persona, Assistive Technology Navigation Persona
aria-command-name	serious	173	university	Reading Clarity Persona, Assistive Technology Navigation Persona
heading-order	serious	173	healthcare, municipality, university	Task Guidance Persona, Reading Clarity Persona, Assistive Technology Navigation Persona
label-content-name-mismatch	serious	154	healthcare, municipality, university	Task Guidance Persona, Reading Clarity Persona, Assistive Technology Navigation Persona

landmark-no-duplicate-banner	moderate	151	healthcare, university	Reading Clarity Persona, Assistive Technology Navigation Persona
aria-allowed-role	serious	129	healthcare, university	Reading Clarity Persona, Assistive Technology Navigation Persona
aria-allowed-attr	critical	116	healthcare, university	Reading Clarity Persona, Assistive Technology Navigation Persona
target-size	serious	91	healthcare, municipality, university	Task Guidance Persona, Assistive Technology Navigation Persona
image-redundant-alt	serious	71	municipality, university	Task Guidance Persona, Reading Clarity Persona, Assistive Technology Navigation Persona
image-alt	critical	67	university	Task Guidance Persona, Reading Clarity Persona, Assistive Technology Navigation Persona
meta-viewport-large	minor	31	healthcare, municipality	Reading Clarity Persona, Assistive Technology Navigation Persona
landmark-banner-is-top-level	moderate	30	university	Reading Clarity Persona, Assistive Technology Navigation Persona
nested-interactive	serious	30	healthcare, municipality, university	Task Guidance Persona, Assistive Technology Navigation Persona
presentation-role-conflict	minor	30	healthcare	Reading Clarity Persona, Assistive Technology

				Navigation Persona
scrollable-region-focusable	serious	29	university	Task Guidance Persona, Assistive Technology Navigation Persona
list	serious	28	healthcare, municipality, university	Task Guidance Persona, Reading Clarity Persona, Assistive Technology Navigation Persona
page-has-heading-one	moderate	27	healthcare, municipality, university	Task Guidance Persona, Reading Clarity Persona, Assistive Technology Navigation Persona
label	critical	26	university	Task Guidance Persona, Reading Clarity Persona, Assistive Technology Navigation Persona
landmark-one-main	moderate	24	healthcare, municipality, university	Reading Clarity Persona, Assistive Technology Navigation Persona
button-name	critical	18	municipality, university	Task Guidance Persona, Reading Clarity Persona, Assistive Technology Navigation Persona
document-title	serious	18	university	Task Guidance Persona, Reading Clarity Persona, Assistive Technology Navigation Persona
empty-heading	minor	18	healthcare, municipality, university	Task Guidance Persona, Reading Clarity Persona, Assistive Technology Navigation Persona
html-has-lang	serious	18	university	Task Guidance Persona, Reading

				Clarity Persona
link-in-text-block	serious	17	healthcare, municipality	Task Guidance Persona, Reading Clarity Persona, Assistive Technology Navigation Persona
landmark-complementary-is-top-level	moderate	16	healthcare, university	Reading Clarity Persona, Assistive Technology Navigation Persona
frame-title	serious	12	healthcare, municipality	Task Guidance Persona, Reading Clarity Persona, Assistive Technology Navigation Persona
aria-valid-attr	critical	10	healthcare, municipality	Reading Clarity Persona, Assistive Technology Navigation Persona
label-title-only	serious	7	healthcare, university	Task Guidance Persona, Reading Clarity Persona, Assistive Technology Navigation Persona
aria-valid-attr-value	critical	6	municipality	Reading Clarity Persona, Assistive Technology Navigation Persona
landmark-main-is-top-level	moderate	6	university	Reading Clarity Persona, Assistive Technology Navigation Persona
landmark-no-duplicate-main	moderate	6	university	Reading Clarity Persona, Assistive Technology Navigation Persona
select-name	critical	5	healthcare, university	Task Guidance Persona, Reading Clarity Persona, Assistive Technology Navigation Persona

accesskeys	serious	4	municipality	Task Guidance Persona, Assistive Technology Navigation Persona
aria-input-field-name	serious	4	university	Reading Clarity Persona, Assistive Technology Navigation Persona
skip-link	serious	4	municipality	Task Guidance Persona, Reading Clarity Persona, Assistive Technology Navigation Persona
role-img-alt	serious	3	municipality	Task Guidance Persona, Reading Clarity Persona, Assistive Technology Navigation Persona
aria-hidden-focus	serious	2	municipality	Task Guidance Persona, Reading Clarity Persona, Assistive Technology Navigation Persona
listitem	serious	2	healthcare	Task Guidance Persona, Reading Clarity Persona, Assistive Technology Navigation Persona
landmark-contentinfo-is-top-level	moderate	1	healthcare	Reading Clarity Persona, Assistive Technology Navigation Persona
landmark-no-duplicate-contentinfo	moderate	1	healthcare	Reading Clarity Persona, Assistive Technology Navigation Persona
table-fake-caption	serious	1	municipality	Task Guidance Persona, Reading Clarity Persona, Assistive Technology Navigation Persona

td-has-header	serious	1	healthcare	Task Guidance Persona, Reading Clarity Persona, Assistive Technology Navigation Persona